WEST UNIVERSITY OF TIMIŞOARA
FACULTY OF COMPUTER SCIENCE
MASTER STUDY PROGRAM: INTELLIGENT SOFTWARE ROBOTICS

MASTER THESIS

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TIMIŞOARA
2026

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AffectLab: Design and Evaluation of an Emotion-Aware Multimodal System for Adaptive Difficult-Conversation Rehearsal

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Abstract

Practising difficult interpersonal conversations can be inaccessible, inconsistent, or uncomfortable, while conventional conversational systems often ignore context, vocal cues, and uncertainty. This dissertation presents AffectLab, a non-clinical application for adaptive communication rehearsal that combines contextual text and speech emotion recognition with deterministic dialogue, feedback, safety, and fallback policies. Emotion models were evaluated on five speaker-independent IEMOCAP folds using classification, calibration, and dialogue-level bootstrap measures; the integrated application was assessed through authored cases, automated and full-stack tests, accessibility checks, restart recovery, and local latency measurements. Three-turn context increased text macro F1 from 0.6651 to 0.7332. Equal-weight text–audio fusion reached 0.7883 macro F1 and 0.7832 accuracy, improving macro F1 over contextual text by 0.0551 (95% interval 0.0407–0.0713), although its ECE of 0.1619 exposed a calibration trade-off. The application passed 245 backend tests, 76 desktop/mobile browser cases, production frontend checks, and two database-backed restart journeys; deterministic conversation turns had a 5.836 ms 95th-percentile local latency. In a supplementary single-arm pilot, 12 of 15 enrollees completed the protocol, while realism and feedback usefulness each had a median rating of 5/7. Required pilot tasks bypassed multimodal inference and affect-aware pacing, so these ratings describe only the baseline rehearsal workflow and provide no evidence about the acceptability or benefit of emotion-aware adaptation. The small, uncontrolled pilot also cannot establish causal benefit or real-world transfer. Within these limits, AffectLab demonstrates an implementation-complete, reproducibly evaluated, uncertainty-aware system whose suitability for controlled deployment depends on target-environment validation.

Keywords: affective computing; emotion recognition; multimodal fusion; conversational context; difficult-conversation rehearsal; human–computer interaction.

Statement of Scientific integrity

I declare that this MSc thesis is an original report and represents my own work, conducted under supervision of Lect. Dr. Todor Ivașcu. All research methods, data collection, analysis and interpretation presented in this thesis are accurately reported and all sources have been cited appropriately where I have made use of ideas and/or visuals of others.

I used generative AI during the writing and research process. The details are in the Appendix A.

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List of Abbreviations

AI	Artificial intelligence
API	Application programming interface
BERT	Bidirectional Encoder Representations from Transformers
BSON	Binary JSON
CI	Continuous integration
CORS	Cross-origin resource sharing
CPU	Central processing unit
CSV	Comma-separated values
DOM	Document Object Model
ECE	Expected calibration error
ERC	Emotion recognition in conversation
F1	Harmonic mean of precision and recall
GDPR	General Data Protection Regulation
GPU	Graphics processing unit
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IEMOCAP	Interactive Emotional Dyadic Motion Capture
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large language model
NLL	Negative log-likelihood
P95	95th percentile

PCM	Pulse-code modulation
RQ	Research question
SD	Standard deviation
SER	Speech emotion recognition
SHA-256	Secure Hash Algorithm 256-bit
SMTP	Simple Mail Transfer Protocol
TLS	Transport Layer Security
TTL	Time to live
UA	Unweighted accuracy
UI	User interface
WA	Weighted accuracy
WAV	Waveform Audio File Format
WCAG	Web Content Accessibility Guidelines

Chapter 1

Introduction

1.1 Context and Problem

Difficult interpersonal conversations arise in ordinary situations such as negotiating workload, maintaining a personal boundary, or expressing a relationship need. Although a person may know what they want to communicate, pressure and disagreement can make it harder to state a clear request, respond constructively, or reflect on alternative wording. Structured role-play offers a way to practise these skills before a real interaction, particularly when it provides clear objectives, resistance, reflection, and feedback.

Conventional practice may require another person, a facilitator, or a scheduled and potentially uncomfortable session. A digital system can instead provide private, repeatable access and preserve the same scenario for comparison between attempts. Language models can add conversational variety, but unrestricted generation is not a sufficient foundation for dependable practice: a model may leave its assigned role, vary the intended difficulty, or produce feedback unsupported by the user's words. Scenario progression, outcomes, assessment, and safety therefore require explicit control.

A second challenge concerns affect. Meaning can depend on preceding dialogue and on acoustic delivery as well as the current transcript, which motivates contextual and multimodal emotion recognition. Yet a benchmark label is not a direct measurement of a person's internal state. Predictions can be wrong, overconfident, or inconsistent across text and audio, particularly outside the data on which the models were evaluated. An application must therefore decide not only how to infer affective information, but when that information is sufficiently reliable and appropriate to influence an interaction.

This dissertation addresses the integration problem created by these requirements: how to combine contextual text and speech models with flexible role-play while keeping dialogue outcomes, feedback, safety, privacy, and failure handling predictable. AffectLab treats affect predictions as bounded evidence for modest and reversible pacing choices, not as authoritative psychological judgements. Deterministic policies remain responsible for scenario state, observable communication features, and safety precedence, while optional generative services may realise an already constrained response. Model performance, application conformance, and supporting user feedback are evaluated separately so that evidence from one level is not used to justify claims at another.

1.2 Aim and Objectives

The aim of this dissertation is to design, implement, and evaluate AffectLab, an emotion-aware system for rehearsing difficult interpersonal conversations. The system combines structured role-play, contextual text and audio-based affect recognition, uncertainty-aware adaptation, and evidence-grounded feedback while preserving transparency, user control, privacy, safety, and deterministic fallback behaviour.

The objectives are:

1. to establish, from relevant research, the design constraints for structured communication rehearsal, contextual and multimodal affect recognition, calibrated uncertainty, and responsible human–AI interaction;
2. to develop and compare utterance-only text, contextual text, audio, and late-fusion models under a speaker-independent protocol, evaluating classification and probability quality as distinct concerns;
3. to design and implement scenario-specific dialogue policies and a bounded adaptation policy in which model confidence, modality agreement, availability, user preference, and safety determine only permitted, inspectable actions;
4. to provide evidence-linked feedback and reflective functions within a privacy- and safety-conscious application whose core workflow survives optional-service failure; and
5. to evaluate model behaviour and integrated application quality with evidence appropriate to each claim, while reporting the small user pilot only within its non-adaptive scope.

1.3 Research Questions

The dissertation is guided by four research questions:

RQ1. Does incorporating preceding conversational turns improve text-based emotion recognition compared with classifying each utterance independently?

RQ2. Does combining contextual textual information with acoustic information improve emotion-classification performance compared with the individual text and audio modalities, and what trade-off exists between classification performance and confidence calibration?

RQ3. How can uncertain multimodal affect predictions be integrated into a structured role-play dialogue policy while keeping the system's behaviour explainable, controllable, safe, and robust to disagreement or inference failure?

RQ4. To what extent does the completed AffectLab application satisfy its functional, reliability, recovery, accessibility, responsiveness, and deployment requirements under reproducible engineering evaluation?

1.4 Approach and Contributions

AffectLab uses a layered architecture that separates affect inference, dialogue control, response realisation, feedback, and safety. The experimental pipeline compares an utterance-only text classifier with a model that adds preceding causal dialogue turns. A separate speech model provides acoustic evidence, and late fusion combines the modality distributions. Classification metrics establish predictive performance, while calibration measures assess whether confidence is suitable for operational thresholds. Speaker-independent session folds and paired dialogue-level comparisons keep the principal model comparisons aligned.

Within the application, a deterministic controller represents each scenario as stages, character behaviour, completion conditions, and unresolved outcomes. It selects the next conversational action from observable evidence rather than allowing a language model to decide whether the user has succeeded. An optional generative service may phrase that action more naturally, but validation and deterministic templates preserve the recorded decision when generation is unavailable or unsuitable.

Multimodal estimates enter the interaction through a restricted affect policy. Confidence, agreement between modalities, inference availability, eligibility, and an explicit user preference determine whether the system preserves baseline behaviour, offers a pacing choice, or permits a brief acknowledgement. Safety interruption takes precedence and does not depend on the emotion classifier. This separation also prevents affect predictions from determining scenario success, feedback scores, or participation decisions.

Feedback is derived from inspectable language features associated with each scenario, such as a clear request, maintained boundary, non-blaming statement, acknowledgement, or practical next step. The relevant conversational turns remain available as evidence. Replay, retry, branching, comparison, personal takeaways, and action cards support reflection on alternative responses rather than reducing practice to a single score.

Evaluation follows the same separation of responsibilities. The model study addresses context, modality complementarity, and calibration. Authored policy cases and automated checks address dialogue transitions, fallbacks, persistence, recovery, accessibility, responsive behaviour, and the complete application workflow. Local timing measurements characterise only the tested deterministic path. A small pilot supplies supplementary feedback on the baseline rehearsal workflow; because its required tasks did not exercise multimodal adaptation, it is not evidence that adaptation benefits users.

The dissertation makes five principal contributions:

1. **A controlled evaluation of causal conversational context and multimodal complementarity.** Utterance-only text, contextual text, speech, and fused configurations are compared on matched speaker-independent data, with dialogue-level uncertainty estimates.
2. **An explicit separation between classification performance and operational confidence.** Predeclared fusion and validation-fitted calibration expose the trade-off between the strongest classifier and the probability estimates used by an interactive policy.

3. **A bounded architecture for affect-aware role-play.** Deterministic dialogue and safety logic remain authoritative, while uncertain affect estimates and optional generation are limited to inspectable, reversible actions with defined fallbacks.
4. **Evidence-grounded rehearsal and reflection.** Scenario-specific outcomes and feedback are connected to observable turns, and replay and branching allow alternative responses to be compared from the same conversational context.
5. **A layered verification framework for an integrated application.** Model evaluation, policy conformance, application verification, local performance, and supplementary user feedback are kept distinct, making the scope of each resulting claim explicit.

Together, these contributions concern the responsible integration of affective models into an implemented communication-practice system. They do not imply that AffectLab identifies a user's true emotion or that completing a rehearsal improves a real relationship.

1.5 Scope and Dissertation Structure

AffectLab is intended for adults rehearsing ordinary difficult conversations. The implemented scenarios cover workload negotiation, personal boundary-setting, and expression of a relationship need. Text interaction is primary and voice input is optional. The affective component is limited to supervised categorical recognition from text and audio; it excludes video, physiological signals, multilingual inference, diagnosis, therapy, autonomous crisis assessment, and consequential decisions about another person. Emotion estimates are uncertain application inputs, not findings about an individual's internal state.

The empirical model claims are limited to the defined IEMOCAP protocol. Its ten actors, controlled English-language interactions, annotation process, and categorical labels do not establish generalisation across cultures, accents, devices, populations, or spontaneous application use [BBL+08]. Deterministic language rules are reproducible but cannot recognise every appropriate paraphrase and are not validated measures of general communication ability or psychological constructs. Optional providers introduce additional privacy and reliability boundaries, so the text-based fallback remains part of the intended system rather than an exceptional mode.

Application evaluation establishes behaviour only for the tested cases and configurations. It does not replace independent security review, target-environment load and recovery testing, live-provider evaluation, participatory accessibility work, or validation of trained inference under deployment conditions. The small, uncontrolled pilot concerns the non-adaptive baseline workflow and cannot establish causal benefit, sustained skill development, or transfer to real conversations. AffectLab is therefore presented as an implementation-complete, deployment-oriented research application whose wider use remains conditional on further technical, ethical, and human evaluation.

Chapters 2–6 cover the literature, architecture, implementation, evaluation, and conclusions, respectively.

Chapter 2

State of the Art

Emotion-aware communication rehearsal combines several research traditions whose evidence should not be conflated. Communication-training studies can establish the value of structured practice, conversational-agent research can inform interaction design, and affective-computing benchmarks can measure agreement with operational emotion labels. None of these alone demonstrates that an integrated application is educationally effective, reliable for a new speaker, or safe. This chapter therefore reviews only the literature needed to motivate the research questions and design choices. It distinguishes throughout between predicting a dataset label, controlling application behaviour, and improving a person's communication skills.

2.1 Communication Rehearsal, Role-Play, and Feedback

Role-play turns interpersonal knowledge into observable decisions: a learner must formulate a request, respond to resistance, clarify an intention, and decide whether to maintain or revise a position. Compared with a consequential real interaction, a simulation permits mistakes, repetition, and reflection. Its value nevertheless depends on its learning objectives and feedback, not simply on conversational realism. In a randomised study of 103 medical students, Bosse et al. found that peer role-play and standardised-patient training improved communication performance and self-efficacy relative to a control condition [BSN+12]. A systematic review of 14 virtual-patient studies similarly associated stronger interventions with preparation, scaffolding, debriefing, reflection, and feedback, while finding heterogeneous evidence for comparative effectiveness [LKK+20]. These clinical results support structured rehearsal but do not establish equivalent transfer to everyday workplace or relationship conversations.

Language models can broaden the range of situations available for rehearsal. IMBUE combines personalised interpersonal simulations with just-in-time, framework-based feedback; its randomised evaluation with 86 participants reported gains in measures including self-efficacy and skill mastery, but transfer to new and more difficult situations was more limited than improvement within trained situations [LSR+24]. The result is important because it treats an LLM interaction as a learning activity rather than an open-ended chatbot. It also shows why performance inside a familiar scenario, perceived realism, immediate confidence, and transfer are

separate outcomes.

Feedback is most actionable when it relates a learning goal to observable performance and a feasible next step [HT07]. In communication rehearsal, evidence might include whether the learner made a specific request, gave a reason, acknowledged another perspective, or maintained a stated boundary. Automated evidence remains a proxy: a lexical or rule-based detector can miss a valid paraphrase or misread quotation, negation, and sarcasm, whereas a fluent generated explanation can sound authoritative without being valid. Traceable feedback should therefore identify the turn that triggered an assessment and permit review, retry, or comparison of alternative continuations. Such mechanisms support reflection, but their presence does not prove improvement outside the system. This literature motivates defined scenario goals, legitimate unresolved outcomes, repeatable attempts, and evidence-linked feedback, while requiring any learning claim to be evaluated independently.

Repeatability creates an additional evaluation risk: learners may discover the phrases rewarded by one scenario without acquiring a strategy that survives changed wording, resistance, or context. Evidence recorded inside the simulator should therefore be distinguished from independent assessment and from transfer to an unfamiliar conversation. Self-reported confidence can describe a participant's immediate experience, but without a suitable comparator it cannot identify whether practice, feedback, novelty, or expectation produced the change. This distinction is particularly important for short pilots, which are useful for finding workflow problems and estimating acceptability but rarely support durable learning claims.

2.2 Deterministic and Generative Conversational Agents

Classical task-oriented dialogue systems make conversational state and action selection explicit. RavenClaw, for example, separates domain-specific task logic from reusable dialogue-management capabilities such as turn-taking and recovery [BR09]. For rehearsal, explicit states can represent raising an issue, clarifying a concern, responding to resistance, and reaching either a completed or unresolved outcome. Their principal benefit is controllability: transitions and permissible endings can be explained and tested. Their costs are policy authoring, incomplete coverage of natural phrasing, and repetitive surface responses.

Large language models provide more flexible linguistic realisation and can respond to inputs not anticipated in a template. Research on LLM role-play shows how personas, dialogue goals, and recovery strategies can structure multi-turn simulation [TSG25]. Flexibility is not the same as control, however. In a comparison of generated and human-authored professional role-play responses, Lu et al. found that judged quality deteriorated across turns in the evaluated setting, particularly for naturalness and context maintenance [LJG25]. A plausible reply can also damage a rehearsal exercise by dropping a scenario constraint, becoming prematurely agreeable, or declaring success before its conditions have been met.

A hybrid therefore offers a defensible division of responsibility: deterministic policy selects the stage, resistance level, completion status, and allowed response plan; optional generation varies wording within that plan; and a deterministic re-

sponse remains available if generation fails. Related work on emotionally conditioned and empathetic generation shows that affect-related signals can influence perceived response quality [ZHZ+18, RSLB19], but perceived empathy neither verifies the inferred emotion nor establishes an appropriate instructional action. In a practice scenario, scepticism or refusal may be intentional. Affect-related adaptation should consequently remain subordinate to scenario goals, user choice, and independent safety rules.

2.3 Contextual Text and Speech Emotion Recognition

Emotion recognition requires an operational target. Categorical systems assign labels such as anger, happiness, sadness, and neutral, whereas dimensional accounts describe affect using quantities such as valence and arousal [Ekm92, Rus80]. These representations are useful abstractions rather than direct measurements of private experience. Emotion, mood, feeling, expressive behaviour, and an annotator's interpretation are not interchangeable constructs [Sch05]; observable cues also vary with speaker, culture, setting, and communicative purpose. Accordingly, "emotion recognition" here means prediction of annotated expressive categories. It does not license diagnosis, inference of intent, or a claim that the system knows what a user feels.

The Interactive Emotional Dyadic Motion Capture database (IEMOCAP) provides about twelve hours of scripted and improvised dyadic interactions from ten actors in five sessions, with speech, transcripts, other modalities, and emotion annotations [BBL+08]. Its paired modalities and conversational form make it suitable for controlled text–audio comparison. A common four-class task uses anger, happiness (often including excitement), sadness, and neutral, although published work differs in label mapping, preprocessing, splits, and metrics [LWC+22]. Other conversational corpora, including the multi-party television dialogue in MELD and the written exchanges in DailyDialog, provide useful but non-equivalent domains [PHM+19, LSS+17]. A catalogue of datasets adds little to the present argument: the important point is that results are comparable only under compatible protocols, and an acted English corpus with ten speakers cannot establish robustness across populations, devices, languages, or spontaneous deployment speech.

Conversational samples also violate the assumptions behind an utterance-level random split. Adjacent turns share speakers, recording conditions, subject matter, and affective trajectories. Holding out an entire IEMOCAP session prevents the same actors from appearing in training and test data, while a separate session for validation can protect model selection from test leakage. It does not remove the shared studio, task, and annotation conventions. Statistical comparisons should also respect dependence within dialogues rather than treating every utterance as an independent participant.

Protocol comparability extends beyond the split. Merging excitement with happiness, discarding unmatched utterances, balancing classes, or reporting weighted rather than unweighted measures can materially change the apparent result. A strong published score is therefore context for method selection, not a target that can be compared without reconstructing its data definition. The dissertation in-

stead fixes one primary task and uses identical held-out items for the comparisons that answer its research questions.

Pre-trained text encoders such as BERT and RoBERTa provide contextual token representations that can be adapted to classification [DCLT19, LOG+19]. Yet an utterance such as "That's fine" remains ambiguous without its conversational setting. Emotion-recognition-in-conversation models address this limitation through speaker states, dialogue graphs, commonsense features, or representations of a speaker's previous utterances [MPH+19, GMP+19, GMG+20, LL22]. These studies motivate the use of dialogue history, but many offline arrangements are unsuitable for live interaction. A deployable classifier for turn t may use the current utterance and a bounded history,

$$\hat{y}_t = f(u_t, u_{t-1}, \ldots, u_{t-k}),$$

but not future turns or neighbouring gold labels. A bidirectional token encoder can still be used inside this causally assembled input; bidirectionality over its tokens does not imply access to the future conversation. A matched comparison of utterance-only and causal-context models consequently answers a clearer question than an uncontrolled comparison among unrelated architectures.

Speech contributes pitch, intensity, timing, rate, voice quality, and pauses that a transcript omits, but these cues also reflect identity, health, language, microphone, and environment. Compact descriptors such as GeMAPS offer interpretable acoustic baselines [ESS+16]. Self-supervised encoders including wav2vec 2.0, HuBERT, and WavLM instead learn representations from large speech collections [BZMA20, HBT+21, CWC+22]. Their success in speech representation learning is not itself evidence of emotion-classification accuracy, and a downstream model can still exploit speaker or recording artefacts. Speaker-independent evaluation is therefore essential. At application level, audio must also be optional and recoverable because permissions, noise, silence, short recordings, and the identifying content of a voice introduce failure and privacy costs absent from transcript-only inference.

2.4 Multimodal Fusion, Calibration, and Uncertainty

Text and speech provide complementary but differently structured evidence. Multimodal research distinguishes representation, alignment, and fusion as separate challenges [BAM19]. Early and cross-modal methods can model fine-grained interactions; for example, the Multimodal Transformer uses directional attention between unaligned streams [TBL+19]. Such capacity also increases coupling and makes missing inputs and decision provenance harder to handle. For the present research questions, decision-level fusion supplies a more transparent test. If $p_T(y)$ and $p_A(y)$ are independently produced text and audio distributions, their mixture is

$$p_F(y) = \lambda p_T(y) + (1 - \lambda) p_A(y), \qquad 0 \leq \lambda \leq 1.$$

Equal weighting predeclares $\lambda = 0.5$; a fitted alternative must estimate it using validation data rather than the final test set. The rule is modular, exposes modality disagreement, and permits text-only fallback. It is less expressive than cross-modal attention and can still become confidently wrong when both branches exploit misleading cues.

Classification quality and probability quality answer different questions. Accuracy and macro F1 assess predicted labels, while calibration asks whether confidence corresponds to observed correctness under the evaluated distribution. Neural classifiers are often miscalibrated; temperature scaling can adjust logit magnitudes using a held-out validation set without changing the usual model's top-class ordering [GPSW17]. More flexible multiclass methods such as Dirichlet calibration are possible, but require sufficient validation evidence and can overfit [KPNK+19]. Expected calibration error offers a useful summary but is bin-dependent, so it should be interpreted beside proper scoring measures or reliability evidence rather than as a complete guarantee.

Fusion makes this separation particularly important: the configuration with the strongest classification score need not have the best calibrated confidence. Training, weight selection, calibration fitting, and testing must therefore remain separated inside each speaker-held-out fold. Moreover, calibration is distribution-dependent and can degrade under shift [OFR+19]. A confidence threshold derived from IEMOCAP is an operational presentation rule, not a psychological boundary. Modality agreement is likewise informative but not proof. These limitations support restrained application behaviour: uncertain, conflicting, or unavailable evidence should lead to baseline dialogue or an explicit pacing choice, not to an imposed emotional interpretation.

2.5 Research Gap and Design Implications for AffectLab

Prior work establishes the components of the problem but not their complete evidential chain. Communication-training research supports structured practice, reflection, and feedback; contextual and multimodal models can improve benchmark label prediction; calibration research qualifies the meaning of confidence; and generative agents provide flexible dialogue with known multi-turn control risks. What remains is an integration-and-evaluation gap: a bounded rehearsal system must connect these components without treating benchmark accuracy as emotion understanding, fluent generation as policy compliance, application tests as educational validation, or immediate user ratings as evidence of transfer.

This gap yields four design implications. First, model comparisons require a common, speaker-independent protocol. The contextual text model should be compared with an utterance-only counterpart on the same held-out dialogues, and a separately trained speech branch and transparent fusion rule should be evaluated on matched text–audio cases. Classification and calibration must be reported separately, with test data excluded from model and policy selection. These choices provide evidence for the first two research questions while keeping conclusions specific to the benchmark.

Second, affect inference should be advisory rather than authoritative. Its permitted effects should be few, reversible, and inspectable; scenario progression, completion, feedback, and safety remain governed independently. This boundary is especially important because emotionally expressive input can occur during serious distress. Risks identified for generative wellness applications justify clear scope limits and a safety route that does not depend on a four-class emotion model [DFC24].

In the European legal context, the Artificial Intelligence Act also restricts particular emotion-inference uses in workplace and educational settings, subject to its stated scope and exceptions [Eur24]. Voluntary personal rehearsal is not equivalent to institutional surveillance, but deployment would still require case-specific legal, privacy, and ethical assessment. Explicit microphone activation, minimised audio retention, access control, and withdrawal procedures follow from this boundary rather than from a claim of compliance by design alone.

Third, hybrid dialogue should keep control decisions separate from surface wording. Explicit scenario states and completion rules make intended resistance, unresolved outcomes, safety precedence, and generator fallback testable. Optional generation can improve variation but cannot alter these decisions. Evidence-linked feedback similarly remains an auditable implementation construct, not a validated measure of communication competence. These constraints turn the third research question into a question about specified application behaviour rather than machine empathy or psychological validity.

Finally, evaluation must match each claim. Model experiments assess annotated-label prediction and calibration; authored policy cases assess rule conformance; application tests assess functional, persistence, recovery, accessibility, interface, and performance behaviour; and a limited pilot can describe completion and participant perceptions. Passing one layer cannot substitute for another. In particular, a small, single-arm pilot cannot establish that affect adaptation caused improvement or that learning transfers to real conversations. This layered strategy addresses the fourth research question while preserving the distinction between technical verification and human benefit.

The resulting position is deliberately narrower than a new universal emotion model or an unconstrained role-playing companion. AffectLab investigates whether causal conversational context and text–audio fusion improve a defined speaker-independent task, whether probability quality changes the operational interpretation of those predictions, and whether a complete rehearsal application can apply bounded adaptation while remaining testable under failure. Chapter 3 translates these implications into authority boundaries and requirements; Chapter 4 describes the mechanisms; and Chapter 5 evaluates each evidence layer.

Chapter 3

Proposed System Architecture

3.1 Architecture and Design Principles

AffectLab is a text-first application for rehearsing difficult interpersonal conversations. Its architecture combines deterministic application logic, trained affect-recognition models, optional generative services, and a browser interface, but it is not an end-to-end conversational model. The central design principle is separation of authority: probabilistic components may provide affective evidence or flexible wording, whereas explicit policies retain control of safety, dialogue progression, completion, and feedback evidence. This division makes consequential decisions inspectable and allows the core rehearsal to continue when a model or provider is unavailable.

Figure 3.1 shows the resulting information and authority flow. The interaction layer supports scenario selection, text and optional voice input, conversation display, and reflection. The application layer authenticates the request, validates ownership, coordinates the active session, and persists its state. Within the decision boundary, safety and dialogue policies are authoritative; affect inference supplies bounded evidence to the dialogue policy but cannot replace either authority. Response realisation converts a selected action into text using a constrained generator or its deterministic template. Feedback and reflection operate on recorded policy evidence rather than on unverified generated claims.

The architecture is deliberately text-first. Every conversational turn contains text, either typed by the user or reviewed after optional transcription. Text therefore remains available to the safety and dialogue components even when microphone permission is denied, audio is invalid, trained inference cannot be loaded, or an external provider fails. Raw audio is processed transiently and is not retained as part of conversation history. Optional services can improve modality coverage or linguistic variety, but their loss must not remove the core workflow.

The three standard scenarios—workload negotiation, personal boundary-setting, and expression of a relationship need—share a structural contract. Each defines its objective, stages, difficulty-dependent resistance, observable communication features, response actions, and possible outcomes. A deterministic policy evaluates the current message against that contract and records both the selected action and its evidence. The response-realisation layer receives the action only after this decision; consequently, generated wording cannot redefine success, remove intended resistance, alter the evidence record, or override a safety interruption.

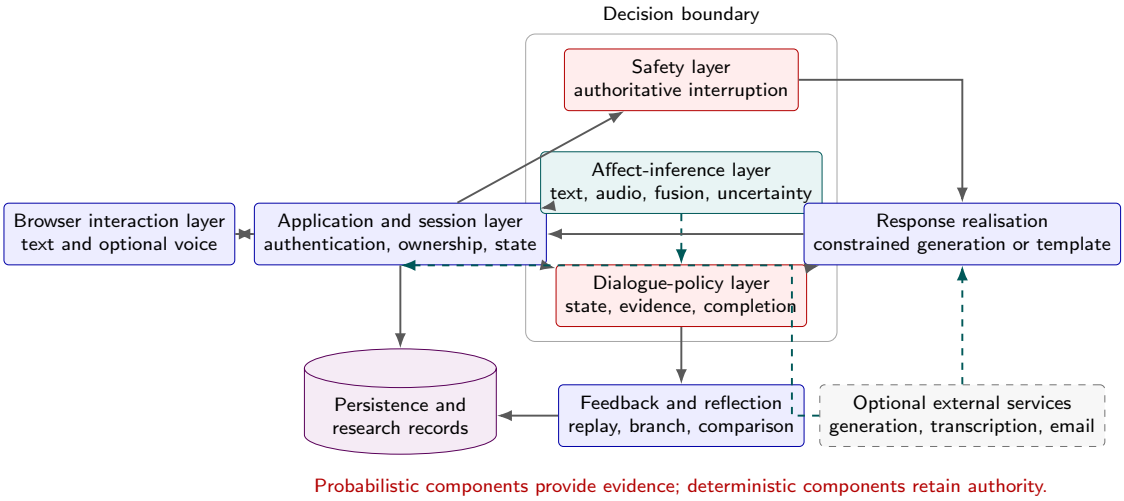


Figure 3.1: Logical architecture of AffectLab and separation between probabilistic evidence and authoritative application control.

Persistence stores the textual turns and enough decision metadata to reconstruct the interaction: state transitions, detected features, selected actions, reason codes, affect-policy decisions, component versions, and generation or fallback provenance. Completed sessions can then support evidence-linked feedback, replay, retry, branching, comparison, personal takeaways, and action cards. Research records form a separate, minimised projection rather than a copy of the participant-facing conversation.

Component boundaries are expressed through narrow contracts rather than shared access to model or provider internals. Inference returns probabilities and provenance; the affect policy returns a permitted adaptation decision; the dialogue policy returns state, evidence, and an action; and response realisation returns wording plus its source. Explicit unavailable and rejected states are part of these contracts. This makes failure behaviour testable, prevents an optional component from silently acquiring authority, and permits models, persistence implementations, or external providers to be replaced without rewriting the scenario policies.

These boundaries align the architecture with the research questions. Contextual text, speech, fusion, and calibration support RQ1 and RQ2; the separation of affect evidence from dialogue and safety authority addresses RQ3; and the integrated, recoverable application provides the object evaluated for RQ4. Model accuracy, policy conformance, application quality, and pilot feedback therefore remain distinct forms of evidence.

3.2 Grouped System Requirements

The original specification contains 29 functional requirements (FR1–FR29) and 14 non-functional requirements (NFR1–NFR14). Table 3.1 consolidates them into nine architectural groups while retaining every identifier used by the implementation and the Chapter 5 traceability analysis. This grouping is sufficient for the design argument; the complete atomic wording is reproducibility material designated for Appendix B during the appendix reorganisation.

Table 3.1: Grouped functional and non-functional requirements.

Group	**Identifiers**	**Required capability or quality**
Identity and data	FR1–FR3, FR26; NFR3–NFR5, NFR13	Authenticated ownership, clear disclosure and preferences, user-controlled optional features, minimised retention, deletion, security controls, and consistent stored state.
Preparation	FR4–FR8, FR19	Scenario preparation and configuration, text-first lifecycle control, and explicit successful, unresolved, manual, maximum-turn, and interrupted outcomes.
Voice and affect	FR9, FR11–FR14; NFR2, NFR14	Reviewed transcription, causal text context, separate audio evidence, declared late fusion, graceful missing-modality handling, and uncertainty-aware bounded adaptation.
Safety	FR10; NFR1	Textual safety checks interrupt ordinary role-play independently of affect classification and before generation.
Dialogue	FR15–FR18; NFR6, NFR8, NFR11–NFR12	Deterministic state transitions and evidence capture, constrained response wording, equivalent fallbacks, auditability, consistency, modularity, and testability.
Feedback	FR20–FR25	Evidence-grounded feedback, controlled phrasing, replay, retry, branching, compatible-attempt comparison, and user-authored reflection.
Research	FR27–FR28; NFR7	Versioned consent and eligibility, withdrawal-aware progress, pseudonymous minimal exports, and reproducible protocol and schema versions.
Support routing	FR29	Transparent lexical routing outside role-play, with tentative rule matches that have no authority over safety, eligibility, scoring, or scenario outcomes.
Operational quality	NFR9–NFR10	Bounded failures and recovery, measurable responsiveness, keyboard and assistive-technology support, visible status, reduced motion, and responsive layouts.

The groups also distinguish core from optional capability. Authentication, text interaction, scenario policy, safety handling, deterministic response templates, evidence-based feedback, and session management form the core. Voice capture, transcription, trained multimodal inference, generative wording, and research participation are optional. A valid implementation may expose an optional feature only when its configuration and failure path preserve the core requirements.

Consolidation does not weaken the atomic acceptance criteria. Each identifier still denotes a separately testable obligation, while a group states the common architectural boundary through which those obligations are implemented. Chapter 5 therefore traces identifiers to evidence rather than treating a group as a single pass/fail claim. This is particularly important for optional features: successful text fallback does not prove trained-inference quality, and successful provider generation does not replace verification of the underlying deterministic action.

3.3 Dialogue, Affect Adaptation, and Safety Boundaries

Each turn follows an authority-ordered pipeline. Authentication, ownership, input, and stale-state checks occur first. The safety layer then examines the text before affect adaptation or response generation. If a supported crisis-related expression

is detected, ordinary role-play stops, a predefined response is returned, and the interruption is recorded. Safety does not use the four-class emotion classifier: a neutral prediction cannot cancel explicit textual evidence, and a generator cannot suppress the interruption.

When no interruption is required, contextual text inference uses the current utterance and a fixed causal window of preceding turns. If valid speech is available, the audio model produces a separate distribution; calibrated modality outputs may then be combined through the declared late-fusion configuration. Availability, confidence, agreement, and component versions remain visible to the policy. Missing audio is absence of evidence, not evidence about the user's affect.

The affect policy maps this information to a small set of reversible actions. Depending on eligibility, user preference, dialogue state, confidence, and modality agreement, it preserves baseline behaviour, offers a pacing choice, or permits a tentative acknowledgement. It cannot determine crisis handling, scenario completion, feedback scores, study eligibility, or research inclusion. Thresholds are operational settings rather than psychological boundaries, and the interface must not present a predicted category as knowledge of an internal state.

In parallel, the scenario policy identifies observable features such as a specific request, maintained boundary, non-blaming statement, acknowledgement, or practical next step. It selects the next action and decides whether the exchange advances, continues under resistance, closes unresolved, reaches a defined outcome, or stops at the turn limit. A generator may phrase the selected action but cannot change it; invalid or unavailable generation invokes the template for the same action. Feedback uses the recorded features and supporting turns, with optional generation limited to restating that authoritative record.

Outside active role-play, deterministic lexical rules may select a bounded support strategy. Labels such as intent, cause, readiness, resistance, or cognitive pattern denote which authored rule fired, not validated psychological constructs. They remain contestable, have no authority over the rehearsal or research workflow, and are evaluated only for rule conformance.

The same authority order governs retries and concurrent requests. A turn is accepted only against the expected session state; a stale update cannot overwrite a newer transition. Retry and branch operations copy declared context but create distinct continuation records, so later feedback can separate inherited turns from new evidence. These constraints make replay reproducible and prevent a delayed provider response from retroactively changing a dialogue decision that has already been persisted.

3.4 Ethics, Privacy, and Design Limitations

AffectLab is intended for voluntary adult rehearsal of ordinary interpersonal conversations. It is not a diagnostic, therapeutic, crisis-monitoring, employment-assessment, educational-assessment, or surveillance system. Completing a scenario means satisfying its declared simulated conditions; it does not establish that the response is appropriate in every real relationship or that the user has improved a general communication skill.

The most important ethical constraint is that annotated expression is not di-

rect access to emotion. IEMOCAP contains ten actors in controlled English-language interactions, and its categories simplify ambiguous affective experience [BBL+08]. Speaker-independent evaluation limits identity leakage but does not establish demographic, cultural, device, or deployment validity. Confidence and text–audio agreement also do not prove correctness, especially under distribution shift [GPSW17, OFR+19]. The architecture therefore confines predictions to low-consequence choices, preserves a non-adaptive path, and allows the user to decline voice or adaptation without penalty.

Privacy constraints follow purpose limitation and data minimisation [Eur16]. Microphone use requires an explicit action; transcripts are reviewable before submission; raw audio is not intentionally retained; conversation storage is time-bounded and account-owned; and research exports exclude direct identifiers, raw conversations, audio, feedback prose, custom scenario wording, and personal takeaways. Pseudonymous records remain potentially linkable while withdrawal is supported and must not be described as anonymous. External transcription or generation also creates a provider boundary that requires disclosure, protected credentials, and a text-only fallback.

Safety remains deliberately limited. Conservative English lexical checks can miss indirect, coded, misspelled, sarcastic, or multilingual expressions and can interrupt quoted or hypothetical language. The application does not know the user's location, provide continuous monitoring, perform clinical triage, or contact emergency services. These limits must accompany the interruption response and prevent the application from being presented as a substitute for professional support.

Fairness and accessibility also shape the design. Text must remain a complete route through the application so that an accent, transcription error, unavailable microphone, or preference not to speak does not remove core functionality. Keyboard operation, visible focus, labelled controls, status announcements, reduced-motion support, and responsive layouts are architectural requirements rather than interface polish. Even so, aggregate model scores and automated interface checks cannot establish equitable performance or accessibility across cultures, communication styles, disabilities, and devices; representative evaluation remains necessary.

These safeguards are design constraints, not evidence of legal compliance. Processing purpose, institutional role, jurisdiction, provider terms, and recipients of affect outputs can change the applicable obligations. In particular, voluntary rehearsal does not authorise employers or educational institutions to use individual emotion inferences for assessment or discipline [Eur24]. Any such change would require a new ethical and legal analysis rather than reuse of the present evaluation.

Deterministic dialogue and feedback improve reproducibility but recognise only anticipated states and language patterns; valid paraphrases, indirect communication, and culturally specific wording may be missed. Generative wording remains probabilistic despite validation. Accessibility checks cannot establish usability for every disabled user, and technical safeguards cannot guarantee confidentiality in an incorrectly configured or compromised deployment. The principal empirical limitations—benchmark validity, authored test coverage, local performance scope, and the non-adaptive pilot—are consolidated with the results in Chapter 5. Accordingly, deployment still requires target-environment security, privacy, accessibility, provider, model, and legal review; the dissertation does not claim compliance certification.

Chapter 4

Implementation

4.1 Technology and Application Overview

AffectLab is implemented as a modular web application whose trained models advise, but do not control, the rehearsal workflow. A React and TypeScript client presents preparation, role-play, feedback, and reflection views. A Python and FastAPI backend owns authentication, dialogue state, safety decisions, feedback evidence, and persistence in MongoDB. Contextual text and speech classifiers can add calibrated affect estimates, while optional provider services can transcribe audio or realise an authorised response in natural language. Each optional component is behind a narrow interface, so the text-based deterministic workflow remains available when model artefacts, credentials, or network services are absent.

This separation is also reflected in deployment. The browser submits typed requests and replaces its local view with the version committed by the backend; it does not calculate authoritative transitions or scores. Backend routes validate transport objects, application services implement policy, and a repository interface supplies either MongoDB persistence or an in-memory substitute for isolated demonstrations. Private datasets, model weights, credentials, and transient audio remain outside version-controlled application code. Dependency sets separate the core web service from the larger local-inference runtime, and containers package the compiled client, API, and database without embedding private artefacts.

Typed boundaries carry the distinctions through the application. A conversation turn can retain the emotion distribution available at that instant, the selected support or character action, the response source, and the pacing decision, while a role-play decision additionally retains its policy version and state snapshots. Absence is represented explicitly rather than as a zero-valued observation. This permits later replay and evaluation to distinguish deterministic, provider-backed, model-assisted, and failed paths without reconstructing them from a transcript.

The implementation is text-first and failure-tolerant. Dialogue progression, evidence collection, crisis interruption, feedback, replay, and session management do not require a graphics processor or external provider. Probabilistic outputs pass through a bounded pacing policy. A language model may phrase a response or feedback item, but it cannot choose an action, declare success, create evidence, or override safety. These authority boundaries organise the mechanisms below.

4.2 Emotion-Recognition Pipeline

The model pipeline converts a private IEMOCAP release into separate text and speech classifiers and then packages compatible full-data artefacts for optional application inference. Preprocessing joins annotations, transcripts, and sentence-level waveforms by utterance identifier, rejects duplicates or missing inputs, and records source and manifest digests. The principal task maps anger, happiness plus excitement, neutrality, and sadness to a fixed four-class order, producing 5,531 matched text–audio utterances. Licensed media and row-level derived data are kept outside the repository. An exploratory six-class mapping is retained only as a secondary research configuration and is not used by the deployed multimodal service.

Figure 4.1 distinguishes evaluation from deployment packaging. Five speaker-independent folds are constructed from complete IEMOCAP sessions: one session is held out for testing, one for validation, and three for training. Because each actor occurs in one session, preprocessing can verify that speakers do not cross split boundaries. These folds provide the evidence reported in Chapter 5. Only after the experimental configuration is frozen are final text and audio models trained on all eligible examples for application use; their training outputs are not treated as a new held-out result.

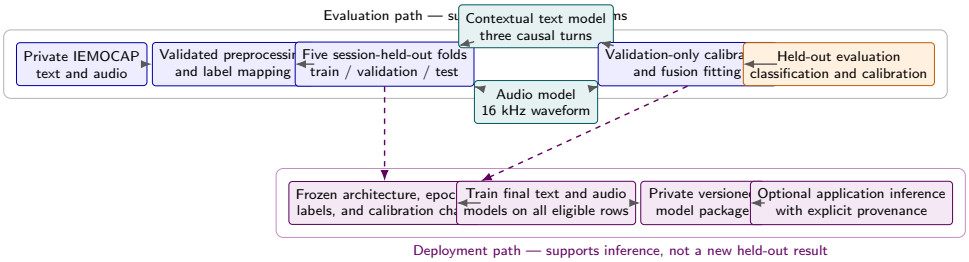


Figure 4.1: Emotion-recognition evaluation and deployment pipeline. Full-data packaging creates application artefacts, not additional held-out evidence.

The contextual text classifier fine-tunes `microsoft/deberta-v3-small`, a compact DeBERTaV3 encoder [HGC21]. Its utterance-only configuration receives the target transcript, whereas the contextual configuration prepends at most three earlier turns from the same dialogue. Each earlier turn is marked as spoken by the same or other speaker relative to the target, and the target receives its own marker. Future turns, emotion annotations, and dialogue outcomes are excluded. The backend reconstructs the same causal format from the stored conversation, avoiding a mismatch between training and inference context.

The acoustic classifier fine-tunes `facebook/wav2vec2-base` on raw waveforms [BZMA20]. Audio is converted to mono, resampled to 16 kHz, limited to 20 seconds, and dynamically padded within a batch. The feature encoder is frozen while the transformer and classification head are trained. Text and audio artefacts share the same label order, and their prediction files must have identical utterance identifiers before fusion is permitted.

Both classifiers are selected by validation macro F1 rather than test performance. Text fine-tuning uses cross-entropy, a small learning rate, warm-up, weight decay, and early stopping; speech fine-tuning adds gradient accumulation and checkpointing to fit the raw-waveform model within available memory. Each fold records

its seed, session assignment, label order, training history, environment, validation calibration, and row-level test distributions. The fusion program verifies these identifiers and reconstructs the saved single-modality metrics before combining probabilities, preventing a result from being produced from mismatched rows or class orders.

Decision-level late fusion keeps the modalities independently trainable and inspectable [BAM19]. For text logits z_t and audio logits z_a, validation-fitted temperatures first produce $p_t = \mathrm{softmax}(z_t/T_t)$ and $p_a = \mathrm{softmax}(z_a/T_a)$. The distributions are then combined as

$$p_m = wp_t + (1 - w)p_a, \tag{4.1}$$

with a predeclared $w = 0.5$ for the principal research comparison. A separate deployment path chooses w and a final temperature from validation-role predictions only. Temperature scaling changes probability concentration without changing the largest logit [GPSW17]; classification and calibration are consequently evaluated as separate properties. The packaged global parameters are operational settings rather than independent performance evidence because their pooled fitting procedure reuses utterance labels across rotating fold roles.

At inference, the backend returns complete calibrated distributions, selected labels, confidence, modality agreement, and separate queue and execution times. Confidence below 0.55 is presented as low, from 0.55 to below 0.75 as moderate, and at least 0.75 as high. These are operational display and policy thresholds, not psychological boundaries. Missing audio, disagreement, invalid distributions, or unavailable models remain explicit; they are never converted into a stronger estimate. Calibration can also deteriorate under shifts in speaker, language, device, and interaction style [OFR+19], so the application treats the distributions as uncertain evidence rather than descriptions of internal state.

The private inference package contains the two models, their tokenizer or feature extractor, the fixed label order, input limits, and the complete calibration chain. A provenance manifest binds these files to source and configuration hashes without publishing licensed weights. Models load lazily and run behind an asynchronous lock to bound memory contention. Multimodal inference is disabled by default and failure returns the exchange to the deterministic text path.

4.3 Backend and Request Processing

The FastAPI backend is the authoritative coordination layer. API routes and Pydantic schemas constrain inputs and outputs, application services coordinate authentication, conversation, model, and provider functions, and repositories encapsulate mutable state. This layering keeps transport logic separate from dialogue policy and permits deterministic or trained implementations to satisfy the same interfaces. Session documents retain ordered turns, role-play snapshots, evidence, policy decisions, feedback, model metadata, version, and expiry time; response schemas expose only what each interface requires.

Authentication and routine account operations use the same service boundary but are not part of the research contribution. Passwords and verification or recovery tokens are stored only as adaptive hashes or digests. Short-lived access tokens

authorise API requests, while rotated refresh tokens use HTTP-only cookies. More importantly for the rehearsal data, every session lookup includes both the session and authenticated owner identifiers. Each write also supplies an expected monotonically increasing version and a request identifier; stale or duplicate submissions return a conflict rather than silently overwriting a committed turn. MongoDB provides durable storage and expiry indexes, while the in-memory repository implements the same contract for controlled local use.

Request schemas enforce message, title, reflection, and encoded-audio limits before service execution. Enhanced exchanges must pair their expected version with a unique request identifier, and received probability vectors are accepted only when all values are finite, bounded, normalised, and consistent with the declared leading label. The backend derives agreement and confidence bands again instead of trusting client-authored summaries. These checks keep a malformed enhancement from contaminating the authoritative session even when ordinary text processing can continue.

Figure 4.2 shows the processing order for a conversation exchange. After validation, authentication, ownership, duplication, and version checks, a local crisis-language detector runs before affect inference or response generation. A safety match interrupts the rehearsal through fixed wording and skips optional processing. Otherwise, eligible requests may obtain a multimodal estimate; failure is recorded as unavailability and does not abort the message. The service then appends the user turn, advances the deterministic controller, records evidence, applies any permitted pacing action, realises the authorised reply, and persists the complete transition atomically. The response contains the committed version and server-owned state so the browser does not need to infer what succeeded.

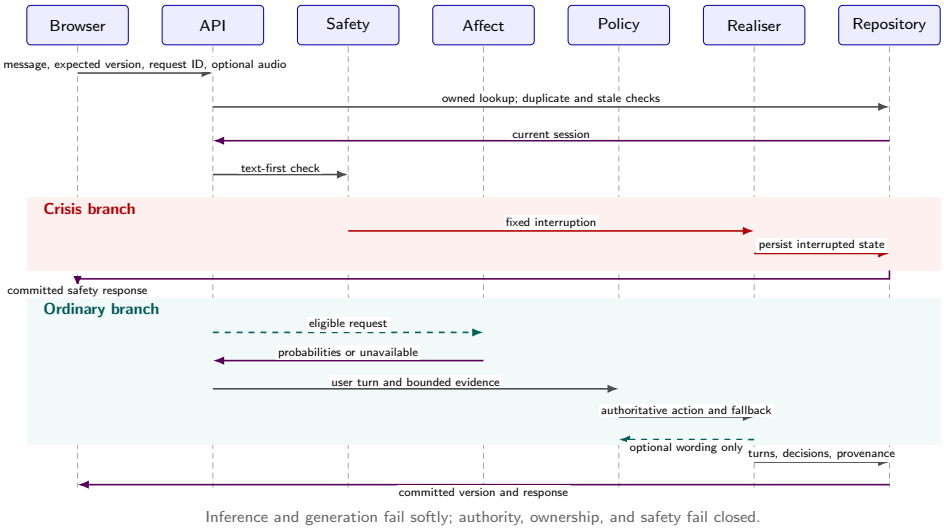


Figure 4.2: Version-bound conversation request with safety precedence and failure-tolerant optional processing.

Provider-backed response realisation follows the same authority rule. It receives the selected support strategy or role-play action and must return a typed, short response consistent with that action. Empty, refused, malformed, unsafe, or contradictory output is discarded in favour of deterministic wording. A small rule-based analyser can expose tentative lexical intent, possible patterns, and authored

support priorities outside active role-play, but these values are rule matches rather than learned probabilities or validated psychological measurements. During role-play, neither these heuristics nor provider output may advance the scenario, create evidence, or determine success.

4.4 Role-Play and Affect-Adaptation Engine

A scenario is represented by its participant objective, character, opening line, supported observable skills, difficulty behaviour, completion conditions, and maximum turn count. Standard scenarios cover workload prioritisation, maintaining a boundary, expressing a relationship need, workplace feedback, deadline negotiation, and shared responsibilities; the first three form the fixed pilot sequence. Custom scenarios reuse only supported features and present any prepared objection as an editable rehearsal prompt, not as a prediction of another person.

The enhanced workload, boundary, and relationship scenarios are implemented as versioned state graphs. Workload practice moves from describing the problem through acknowledging constraints to a workable plan. Boundary practice requires a clear refusal to survive authored pressure before respectful closure. Relationship practice requires a specific need, acknowledgement of perspective, and a practical request. A transition occurs only when the evidence required at the current stage is present; contradictory or unsupported input leaves the controller in place. Attempts can also end at the turn limit, by manual finish, or through safety interruption, and each decision stores its before-and-after snapshot so a restored session does not reconstruct state from prose.

Difficulty controls authored resistance and cooperation rather than changing what a detected feature means. Eligible additional practice can also select a co-operative, rushed, or sceptical character profile, each backed by a versioned policy; profiles are excluded from required pilot tasks. Every processed rehearsal turn records the chosen action, reason codes, linked user and assistant turns, supporting evidence identifiers, generation source, and scenario, policy, and scoring versions. Completion reasons remain distinct, allowing the same persisted representation to support recovery, audit, and fair comparison.

Evidence extraction records bounded language features such as a concrete request, specific detail, I-statement, non-blaming wording, maintained boundary, acknowledgement, or feasible proposal. Each item links to its source turn, and features may accumulate only where the scenario policy permits. The rules account for selected negations and contradictions but cannot cover every indirect, sarcastic, multilingual, or complex expression. A match therefore means that an authored rule observed a feature in this rehearsal; it is not proof of intent, competence, or probable real-world success.

The affect-pacing policy in Figure 4.3 is applied only after the controller has selected an action. Safety, dialogue state, and explicit participant preference take precedence. Automatic adaptation is disabled by default and is limited to active, enhanced, non-required rehearsals. Required pilot tasks, retries associated with those tasks, paused or completed attempts, and unsupported scenarios remain on the baseline path. The participant can independently request unchanged, gentler, or more challenging pacing, so the control remains usable without voice.

When automatic adaptation is eligible, missing audio, inference failure, or an unsupported category produces no addition. Low confidence in any component or disagreement between fused, text, and audio leaders produces a neutral pacing choice. Only when all three labels agree and meet the low-confidence boundary can anger or sadness add the fixed acknowledgement "I hear you."; happiness or neutrality adds nothing. The policy never translates a label into anxiety, resistance, readiness, or a clinical state, and it cannot change stage, evidence, success conditions, or feedback scores. The stored decision includes the selected action, reason, preference, model and policy versions, timing, and distributions, allowing the exact behaviour to be explained after reload.

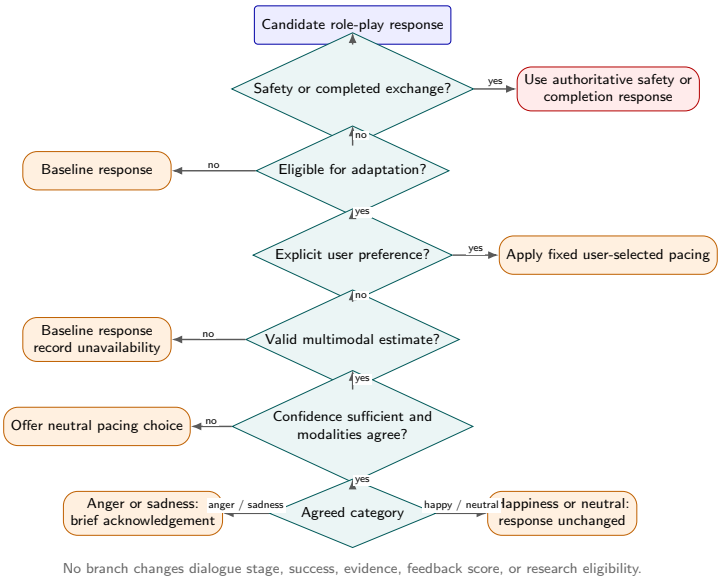


Figure 4.3: Bounded affect-aware pacing. Every terminal action preserves the authoritative dialogue state and evidence.

Response wording begins with the action and fallback supplied by the deterministic controller. If generation is enabled, the provider must repeat the required action, stay in character, avoid coaching language, and preserve any concrete negotiated constraint. Parsing failure, action mismatch, disallowed content, or loss of a material fact restores the fallback. Branches also use deterministic wording so their differences arise from the participant's alternative response rather than uncontrolled regeneration. Safety uses an independent local detector and fixed interruption path before either mechanism; it is an engineering safeguard, not a validated clinical risk assessment.

4.5 Feedback and Frontend Workflow

Feedback is calculated from the evidence and decisions stored during the rehearsal rather than from a new model judgement. Scenario-specific metrics refer to supported features and their source turns; deterministic templates convert them into observed strengths and next-attempt suggestions. The completion reason remains separate, so reaching a turn limit is not relabelled as success and a safety interruption is not scored as a skill failure. Optional provider phrasing may reword

an already calculated item but cannot alter measurements or invent observations. Comparisons with earlier attempts are made only under the same scoring version and are described as differences in observed rehearsal language.

Replay is a read-only projection of the recorded exchanges, state transitions, generation source, linked evidence, and saved pacing decision. Rewind restores a preceding stored snapshot only when the latest user–assistant pair can be removed safely. Branching instead creates a copy-on-write child from a validated pre-turn snapshot; copied context remains distinguishable from the new continuation and is not counted again in feedback. The original attempt is unchanged, and comparison presents both continuations without ranking either as globally better. A personal takeaway and editable action card remain user-authored reflection rather than evaluation evidence.

Measurement boundaries prevent later reflection from entering rehearsal feedback. Missing legacy decisions or model output are displayed as gaps rather than regenerated. A branch validates source ownership, version, policy compatibility, and the selected pre-turn snapshot, then stores its lineage and copied-turn boundary. Action-card suggestions reuse only eligible participant turns inside the measurement window; safety-interrupted content and copied branch context are excluded, and every field remains editable before saving.

The React interface follows the same lifecycle: preparation, active rehearsal, feedback, replay or branching, and an action card. The representative role-play view in Figure 4.4 combines the transcript and composer with scenario progress and optional voice controls. The browser includes the displayed session version and a fresh request identifier, clears a draft only after confirmation, and offers reload after a conflict while preserving unsent text. Semantic controls, visible focus, live status regions, reduced-motion rules, and responsive layouts support keyboard, desktop, and mobile use.

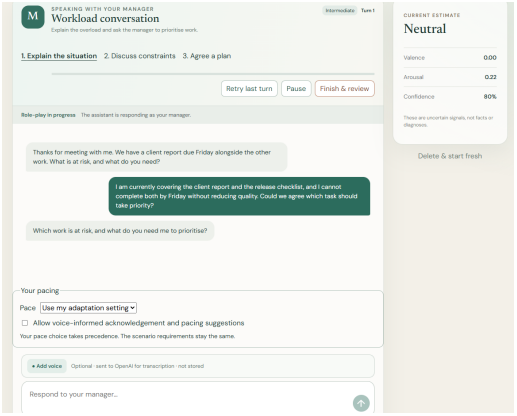


Figure 4.4: Representative role-play workspace with transcript, scenario state, message composer, and optional voice interaction.

Voice input is opt-in for each exchange. The browser requests microphone access only after an explicit action, records at most 20 seconds, converts the signal to mono 16 kHz PCM WAV, and keeps it in component memory. A transcription result enters the composer for review rather than becoming a turn automatically. The reviewed text and optional audio are then sent together so prediction, pacing, and stored wording refer to one exchange. The backend validates the audio and processes it in

memory; raw recordings are not placed in sessions or research exports.

The interface distinguishes the trained multimodal distribution from a separate deterministic lexical display. When multimodal output exists, it shows fused confidence, modality leaders, agreement, the full distribution, timings, and an audio-not-stored notice. Disagreement and unavailable inference remain visible. Reloading uses the persisted decision rather than rerunning the model, preventing a later estimate from being presented as the reason for an earlier response.

Optional research support is limited to a versioned pilot workflow: explicit consent and eligibility, three ordered required tasks, immediate questionnaires, pseudonymous progress records, and a text-free export. Required tasks deliberately bypass multimodal inference and affect-aware pacing. The workflow therefore supports collection for the baseline deterministic rehearsal only; its implementation does not make the pilot evidence about adaptation. This distinction is carried into the evaluation in Chapter 5.

4.6 Security, Persistence, and Deployment

Security controls concentrate on ownership, minimisation, and explicit failure boundaries. MongoDB queries bind session identifiers to their owner; optimistic versions prevent lost updates; production configuration rejects weak signing secrets, wildcard origins or hosts, disabled HTTPS enforcement, and disabled rate limiting. Conversation text persists because continuation, replay, and evidence links require it, but raw audio is transient. Research projections exclude identity, transcripts, free-text feedback, custom wording, and audio and use a retention schedule separate from ordinary sessions. Account deletion cascades through owned operational records, while study withdrawal removes live research records without pretending that irreversibly de-identified frozen exports can still be linked.

Operational and research records have different purposes and expiry fields. The richer account-owned session supports continuation and user-controlled deletion; the pseudonymous study projection carries protocol versions, task classification, questionnaire values, numeric feedback, and event counts without copying conversation content. Freezing creates a checksum-bound de-identified export and closes further protocol mutations. This lifecycle keeps pilot administration from silently expanding the information retained for ordinary practice.

The integrated deployment uses a compiled frontend, FastAPI service, and MongoDB with health checks. Readiness verifies configuration, database access, and required indexes; missing optional transcription or model capability is reported separately and does not make the text-first application unavailable. Expected authentication, validation, rate-limit, provider, model, and concurrency failures become bounded client responses. Authority failures—missing ownership, stale state, invalid consent, or a frozen protocol—fail closed, whereas enhancement failures return to deterministic operation.

Verification evidence is reported once in Chapter 5. Unit, service, browser, and full-stack checks exercise the compiled client and real database while external providers and trained inference remain disabled; performance therefore describes only that controlled local path. Reproduction additionally requires the source revision, locked dependencies, versioned configurations, and private artefact hashes.

Chapter 5

Evaluation and Results

5.1 Chapter Overview

This chapter evaluates AffectLab primarily as a complete technical product. It first reports the speaker-independent emotion-recognition experiments concerning conversational context, acoustic information, multimodal fusion, and calibration. It then evaluates the integrated application through authored engineering cases, automated unit and integration suites, production builds, browser and accessibility checks, persistence and recovery exercises, full-stack smoke journeys, and a reproducible response-time benchmark. A short final subsection reports the supplementary user-pilot results without treating that small study as the principal validation of the application.

These sources support different claims. IEMOCAP results measure classification and probability quality on a controlled corpus. Engineering evidence establishes that the application follows its specified transitions, fallbacks, ownership rules, safety precedence, persistence contracts, responsive layouts, and recovery paths under the tested conditions. These benchmark and engineering evaluations are the principal evidence for the emotion-recognition, adaptation-policy, and application-quality claims. Local timings characterise the deterministic core but are not a substitute for deployment-specific load testing. The small user pilot contributes limited contextual evidence about perceived realism, feedback usefulness, and completion of the baseline rehearsal workflow; because required tasks bypassed multimodal inference and adaptation, it supplies no evidence about the acceptability or benefit of emotion-aware adaptation.

The chapter first maps the research questions to their appropriate evidence. It then presents the common model-evaluation protocol, the contextual-text comparison, the audio and multimodal results, and the classification–calibration trade-off. Application verification, performance results, and the concise pilot summary follow. The final discussion answers each research question only to the extent supported by the available evidence and identifies threats to validity.

5.2 Evaluation Scope and Evidence Hierarchy

5.2.1 Research Questions and Evaluation Levels

Table 5.1 maps the four research questions introduced in Section 1.3 to an evaluation unit and an admissible conclusion. This prevents model accuracy, policy conformance, and integrated application quality from being combined into one undifferentiated claim.

Table 5.1: Mapping between research questions, evidence, and permitted conclusions.

RQ	Evaluation unit	Principal evidence	Permitted conclusion
RQ1	IEMOCAP utterance with causal dialogue history	Utterance-only and three-turn contextual text predictions on identical held-out sessions	Whether causal context improves benchmark text classification.
RQ2	Paired textual and acoustic IEMOCAP samples	Audio baseline, equal-weight fusion, validation-fitted fusion, and calibration metrics	Whether audio adds benchmark classification information and what calibration trade-off follows.
RQ3	Versioned application decision and complete workflow	Authored policy fixtures, automated tests, saved provenance, and full-stack smoke journeys	Whether uncertain estimates are translated into the specified bounded, explainable, and failure-tolerant actions.
RQ4	Integrated application and its principal workflows	Automated suites, desktop/mobile browser checks, production builds, persistence and recovery exercises, full-stack smoke journeys, and local response-time measurements	Whether the completed system satisfies its declared functional and non-functional requirements under reproducible developer-led verification.

RQ1 and RQ2 are empirical machine-learning questions with completed results. RQ3 concerns the bounded integration policy: because the policy is deterministic, controlled input cases can establish whether it implements its specification. RQ4 concerns the quality of the complete implementation and is evaluated through complementary developer-led and automated checks at service, API, persistence, browser, accessibility, recovery, and full-stack boundaries. The small user pilot is reported separately as supporting product feedback rather than as the main answer to RQ4.

5.2.2 Separation of Development, Evaluation, and Deployment Evidence

The model study separates training, validation, and test roles by complete IEMOCAP sessions. Validation predictions select temperatures and, where applicable, fusion parameters; the held-out session supplies evaluation predictions. The predeclared equal-weight fusion is retained as the principal classification comparison because its weight was not selected from the fused test results. Fold-specific validation-

fitted fusion is reported as a calibration analysis rather than substituted retrospectively as the headline classifier.

The final text and audio packages were trained on all 5,531 eligible utterances after the cross-validation design and epoch counts were frozen. These packages have no independent test partition of their own. Their training behaviour and balanced-sample smoke checks are therefore integrity evidence, not model-performance results. Performance claims in this chapter come from the five session-held-out experiments.

Engineering test cases use controlled utterances and probability distributions to exercise known policy branches. Browser and full-stack tests automate test accounts and ratings. Their purpose is to demonstrate repeatability, persistence, recovery, and guard conditions, not to estimate human behaviour or the deployed model's accuracy. The supplementary user pilot is therefore interpreted independently of these tests.

5.3 Emotion-Recognition Evaluation Methodology

5.3.1 Benchmark Task and Speaker-Independent Splits

The primary benchmark uses the four IEMOCAP categories anger, happiness, neutral, and sadness. Excitement is merged with happiness, and utterances carrying other source labels are excluded. The resulting task contains 5,531 matched text–audio examples drawn from 151 dialogues across the five dyadic sessions of IEMOCAP [BBL+08]. The same utterance identifiers and label order are used for the text, audio, and fusion experiments.

Five folds provide speaker-independent coverage. In each fold, one complete session is reserved for testing and a different complete session for validation; the remaining three sessions are used for training. Every session therefore appears exactly once as test data. Keeping both speakers from a dyad within one session prevents an individual's utterances from appearing in the training portion of the fold in which that person is tested.

The utterance-only and contextual text conditions use the same DeBERTa-v3-small architecture, training settings, labels, seed, and folds. Their only intended difference is the input: the contextual condition prepends the previous three causal dialogue turns with speaker-relative markers. The audio condition uses wav2vec 2.0 on the paired 16 kHz waveform. This design supports paired comparisons because competing configurations produce predictions for the same examples.

5.3.2 Classification and Calibration Measures

Classification is reported through accuracy, macro-averaged F1, weighted F1, and per-class F1. Macro F1 gives each class equal weight and is the primary aggregate measure because the class frequencies are unequal. Weighted F1 complements it by weighting each class according to its support. Per-class results are retained so that an aggregate gain cannot conceal deterioration in one category.

Probability quality is considered separately. Negative log-likelihood and the multiclass Brier score are proper scoring rules over the complete predicted distribution. Expected calibration error (ECE) partitions examples by maximum confidence and accumulates the absolute difference between the mean confidence and empirical accuracy within each bin. Because ECE depends on the chosen bins and does not fully describe multiclass calibration, it is interpreted alongside the proper scores and reliability outputs rather than as a complete measure [GPSW17, KPNK+19].

Temperature scaling is fitted on validation logits only. For a logit vector $\mathbf{z}$ and positive temperature T, the calibrated distribution is

$$\hat{\mathbf{p}} = \mathrm{softmax}\left(\frac{\mathbf{z}}{T}\right). \tag{5.1}$$

A single temperature changes confidence concentration but not the predicted class. Fusion weights and a final fusion temperature are also fitted only from validation predictions in the deployment-oriented configurations.

5.3.3 Paired Dialogue-Level Statistical Comparison

Utterances from one conversation are not independent observations. Statistical comparisons therefore use a paired percentile bootstrap over the 151 complete dialogues rather than sampling individual rows. For each of 10,000 replicates, dialogues are sampled with replacement and the same sampled dialogue indices are applied to both compared systems. The difference in accuracy, macro F1, or weighted F1 is then recomputed. The 2.5th and 97.5th percentiles form the reported 95% interval. The procedure uses seed 20260808.

This interval describes uncertainty under resampling of the observed dialogue clusters. It is not a confidence interval over all cultures, microphones, languages, or spontaneous emotional settings. No participant or speaker subgroup inference is attempted because IEMOCAP contains only ten actors and was not designed to support such claims.

5.4 Contextual Text-Emotion Results

5.4.1 Primary Four-Class Comparison

Table 5.2 compares utterance-only text with the three-turn causal context condition. Context improved accuracy from 0.6624 to 0.7304 and macro F1 from 0.6651 to 0.7332. The absolute macro-F1 gain was 0.0681, with a dialogue-bootstrap 95% interval from 0.0504 to 0.0857. All three aggregate intervals excluded zero.

Table 5.2: Four-class contextual-text results on five held-out IEMOCAP sessions.

Metric	Utterance only	Three-turn context	Difference	95% interval
Accuracy	0.6624	0.7304	+0.0680	[0.0496, 0.0864]
Macro F1	0.6651	0.7332	+0.0681	[0.0504, 0.0857]
Weighted F1	0.6640	0.7300	+0.0661	[0.0477, 0.0842]

Table 5.3 reports the complete pooled class-level results underlying the aggregate scores.

Table 5.3: Per-class precision, recall, F1, and support for the four-class text configurations.

Configuration	Class	Precision	Recall	F1	Support
Utterance-only text	Anger	0.7465	0.6863	0.7152	1,103
	Happiness	0.7392	0.6980	0.7180	1,636
	Neutral	0.6134	0.6001	0.6067	1,708
	Sadness	0.5688	0.6827	0.6205	1,084
Three-turn context text	Anger	0.7781	0.6963	0.7349	1,103
	Happiness	0.7989	0.8136	0.8062	1,636
	Neutral	0.6421	0.6388	0.6404	1,708
	Sadness	0.7209	0.7841	0.7512	1,084

Support is the number of held-out examples after the five non-overlapping test sessions are concatenated and is therefore identical for configurations evaluated on the same rows. The per-class differences were not uniform. Happiness F1 improved by 0.0882, neutral by 0.0338, and sadness by 0.1307, with dialogue-bootstrap intervals above zero. Anger F1 improved by 0.0198, but its interval $[-0.0073, 0.0449]$ included zero. The result therefore supports an aggregate benefit from context but does not establish an anger-specific improvement.

The calibrated contextual model obtained pooled ECE 0.0311. This value describes the benchmark distribution after validation-fitted temperature scaling and does not validate the 0.55 application threshold on real AffectLab conversations.

5.4.2 Exploratory Six-Class Comparison

An exploratory task aligned additional source labels to anger, anxiety, frustration, joy, neutral, and sadness. The five held-out sessions contained 7,420 examples: 1,103 anger, 40 anxiety, 1,849 frustration, 1,636 joy, 1,708 neutral, and 1,084 sadness. Thus, anxiety represented only 0.54% of this exploratory dataset, whereas the other classes each contributed more than 1,000 examples. Context again improved all aggregate measures, as shown in Table 5.4. The pooled macro-F1 gain was 0.0732 with an interval from 0.0485 to 0.1009.

Table 5.4: Exploratory six-class contextual-text comparison.

Metric	Utterance only	Three-turn context	Difference	95% interval
Accuracy	0.5674	0.6228	+0.0554	[0.0382, 0.0734]
Macro F1	0.5490	0.6222	+0.0732	[0.0485, 0.1009]
Weighted F1	0.5675	0.6205	+0.0529	[0.0359, 0.0702]

Table 5.5 makes the session-to-session variation explicit. Context produced higher macro F1 in every fold, with fold-specific gains from 0.0475 to 0.0972. The unweighted fold means were 0.5456 and 0.6157; their standard deviations were 0.0330 and 0.0304, respectively. These unweighted means differ slightly from the pooled values in Table 5.4, which recompute macro F1 after concatenating all held-out predictions. The fold counts also expose the fragility of the anxiety result: individual test sessions contained only 2–12 anxiety examples.

Table 5.6 reports the complete class-level metrics for this exploratory task.

Table 5.5: Fold variation in macro F1 for the exploratory six-class comparison. Each fold holds out one complete IEMOCAP session.

Fold	Test n	Anxiety n	Utterance only	Three-turn context	Difference
1	1,377	12	0.5690	0.6180	+0.0490
2	1,357	9	0.5683	0.6655	+0.0972
3	1,535	2	0.4992	0.5813	+0.0820
4	1,519	7	0.5792	0.6268	+0.0475
5	1,632	10	0.5122	0.5870	+0.0748
Mean ± SD	–	–	0.5456 ± 0.0330	0.6157 ± 0.0304	+0.0701

Table 5.6: Per-class precision, recall, F1, and support for the exploratory six-class text configurations.

Configuration	Class	Precision	Recall	F1	Support
Utterance-only text	Anger	0.5857	0.5576	0.5713	1,103
	Anxiety	0.3889	0.5250	0.4468	40
	Frustration	0.5420	0.5165	0.5289	1,849
	Joy	0.7073	0.6779	0.6923	1,636
	Neutral	0.5102	0.4813	0.4953	1,708
	Sadness	0.5004	0.6347	0.5596	1,084
Three-turn context text	Anger	0.6089	0.5476	0.5766	1,103
	Anxiety	0.5778	0.6500	0.6118	40
	Frustration	0.5579	0.5809	0.5692	1,849
	Joy	0.7996	0.8026	0.8011	1,636
	Neutral	0.5174	0.4690	0.4920	1,708
	Sadness	0.6333	0.7408	0.6828	1,084

The class-level effects were heterogeneous. Frustration improved by 0.0402 ($[0.0158, 0.0647]$), joy by 0.1088 ($[0.0804, 0.1371]$), and sadness by 0.1232 ($[0.0926, 0.1557]$). The anxiety F1 difference was comparatively large at 0.1650, but it had a wide interval $[0.0476, 0.3144]$ and arose from only 40 IEMOCAP fear annotations mapped to anxiety. The anger difference of 0.0053 ($[-0.0288, 0.0339]$) and neutral difference of -0.0033 ($[-0.0316, 0.0241]$) both had intervals spanning zero. Consequently, the six-class experiment is not used as the deployed classifier and does not support a strong conclusion about anxiety recognition.

5.4.3 Answer to RQ1

Within the declared IEMOCAP protocol, RQ1 is answered positively. Adding the three preceding causal turns improved the principal text model by approximately 6.8 percentage points in accuracy and macro F1 compared with the utterance-only condition. Pairing, folds, architecture, and optimisation settings were held constant, and the dialogue-level intervals for all aggregate measures excluded zero.

The result is evidence for contextual benchmark classification, not proof that context reveals a user's true emotion. It also does not isolate which part of the representation produced the gain: additional lexical content, speaker markers, or interaction history may each contribute. External validation would be required to determine whether the improvement persists for spontaneous microphone speech and user-authored rehearsal conversations.

5.5 Speech and Multimodal Results

5.5.1 Audio-Only Baseline

The wav2vec 2.0 audio classifier obtained accuracy 0.6279, macro F1 0.6349, weighted F1 0.6229, and calibrated ECE 0.0468. Its complete class-level precision, recall, F1, and support are reported with the fusion configurations in Table 5.9. Audio was weaker than contextual text in aggregate but stronger for anger than the overall pattern might suggest, indicating that the modalities contained different useful information.

The comparison does not imply that the acoustic model measured emotion independently of linguistic content: speech representations can encode phonetic, lexical, speaker, and recording characteristics as well as prosody. It establishes only that the audio branch produced non-redundant predictive information on the matched benchmark task.

5.5.2 Predeclared Equal-Weight Late Fusion

The principal fusion experiment averaged validation-calibrated text and audio posterior distributions with weights of 0.5. Table 5.7 shows that fusion improved accuracy, macro F1, and weighted F1 over contextual text on the same rows. The macro-F1 difference was 0.0551 with a 95% interval from 0.0407 to 0.0713. Every one of the 10,000 dialogue-bootstrap replicates favoured fusion for each aggregate metric.

Table 5.7: Predeclared equal-weight fusion compared with contextual text.

Metric	Context text	Equal fusion	Difference	95% interval
Accuracy	0.7304	0.7832	+0.0528	[0.0381, 0.0683]
Macro F1	0.7332	0.7883	+0.0551	[0.0407, 0.0713]
Weighted F1	0.7300	0.7815	+0.0514	[0.0366, 0.0671]

The complete fusion class-level metrics are reported in Table 5.9. Unlike the contextual-text comparison, the combined system showed substantial performance across all four categories. Because the fusion rule was fixed before inspection of the combined test outcome, this result is retained as the primary evidence that audio contributed complementary benchmark information.

5.5.3 Qualitative Error Analysis

Inspection of matched held-out predictions illustrates both the value and the limits of context and multimodal fusion. IEMOCAP dialogue is not reproduced because of its access conditions, so the examples below are identified by utterance ID. They were selected after evaluation to explain recurring behaviours and are illustrative cases, not estimates of their prevalence.

Context sometimes corrected a locally plausible but wrong reading: for `Ses03M_script01_3_M046` (gold happiness), the utterance-only model predicted neutral, whereas the contextual model predicted happiness. The reverse occurred for

Ses03F_script02_2_M008 (gold neutral), which changed from a correct utterance-only neutral prediction to a contextual anger prediction. Context could also expose rather than resolve uncertainty. For Ses05F_impro08_F028 (gold neutral), the utterance-only model predicted neutral, but the contextual posterior assigned exactly 0.4407 to both happiness and neutral; the stored argmax was happiness. Thus, a changed top label need not represent a decisive contextual interpretation.

Text–audio disagreement was similarly bidirectional. For Ses03M_script02_2_F009 (gold anger), contextual text predicted neutral and audio predicted anger; equal fusion recovered anger. For Ses03F_script02_1_M020 (gold neutral), contextual text was correct but audio predicted anger, and equal fusion narrowly selected anger (0.4555 versus 0.4501 for neutral). Fusion therefore repaired some unimodal errors but could also let a strong wrong modality overturn a correct one. Finally, Ses03F_script01_2_M009 shows a shared failure: both modalities confidently predicted anger for a neutral annotation, so equal fusion reinforced the error. Together with neutral's lowest equal-fusion recall (66.6%), these cases indicate representative failure modes of contextual carry-over, near-tied interpretations, modality dominance, and correlated errors that late averaging cannot correct.

5.5.4 Classification–Calibration Trade-Off

Equal averaging improved classification but produced ECE 0.1619. Separately calibrated modality probabilities therefore did not remain calibrated after fusion. This result is important for AffectLab because confidence, not only the top class, is consumed by the pacing policy. Treating the equal-fusion probability as reliable would have contradicted the evaluation evidence.

Fold-specific validation fitting selected text weights between 0.54 and 0.63 and then fitted a final temperature. This reduced ECE to 0.0283, while macro F1 decreased from 0.7883 to 0.7758. Table 5.8 reports ECE together with negative log-likelihood (NLL) and the multiclass Brier score for every major calibrated configuration whose reported test session was excluded from fitting its parameters. Lower values are better for all three probability-quality measures.

Table 5.8: Classification and probability-quality trade-off for independently evaluated fold configurations.

Configuration	Accuracy	Macro F1	Weighted F1	ECE	NLL	Brier
Contextual text	0.7304	0.7332	0.7300	0.0311	0.7061	0.3845
Audio only	0.6279	0.6349	0.6229	0.0468	0.9441	0.5140
Equal-weight fusion	0.7832	0.7883	0.7815	0.1619	0.6843	0.3619
Fold validation-fitted fusion	0.7717	0.7758	0.7707	0.0283	0.6050	0.3273

The proper scores qualify the ECE-only view. Equal fusion was substantially under-confident, yet its NLL and Brier score were both better than those of either unimodal branch, indicating improvement in the full predictive distribution despite the large confidence–accuracy gap. The validation-fitted fusion achieved the best held-out NLL and Brier score as well as the lowest ECE, despite its modest reduction in classification F1. This agreement across the two proper scores supports its use when probabilities are consumed operationally.

Table 5.9 gives the complete pooled class-level results for the acoustic and multimodal configurations. The repeated support values confirm that all three configurations were evaluated on the same 5,531 held-out utterances.

Table 5.9: Per-class precision, recall, F1, and support for the audio and fusion configurations.

Configuration	Class	Precision	Recall	F1	Support
Audio only	Anger	0.6583	0.7806	0.7142	1,103
	Happiness	0.6343	0.5312	0.5782	1,636
	Neutral	0.6214	0.5363	0.5757	1,708
	Sadness	0.5997	0.7629	0.6715	1,084
Equal-weight fusion	Anger	0.8053	0.8513	0.8277	1,103
	Happiness	0.8221	0.8105	0.8163	1,636
	Neutral	0.7412	0.6657	0.7014	1,708
	Sadness	0.7635	0.8579	0.8080	1,084
Fold validation-fitted fusion	Anger	0.8024	0.7879	0.7951	1,103
	Happiness	0.8208	0.8233	0.8221	1,636
	Neutral	0.7096	0.6710	0.6897	1,708
	Sadness	0.7601	0.8358	0.7961	1,084

The pooled values above are computed after concatenating the utterances from all five sessions, so they are not arithmetic means of the fold scores and sessions with more eligible utterances contribute more observations. Table 5.10 complements them with the primary metric for each held-out session and an unweighted descriptive summary across the five folds. Thus, each session contributes equally to the reported mean and population standard deviation (SD); the range gives the observed minimum and maximum rather than an uncertainty interval.

Table 5.10: Macro F1 by held-out IEMOCAP session, with unweighted variation across the five folds.

Configuration	S1	S2	S3	S4	S5	Mean	SD	Range
Utterance-only text	0.6376	0.7125	0.6443	0.6575	0.6625	0.6629	0.0264	0.6376–0.7125
Three-turn context text	0.6918	0.7556	0.7235	0.7310	0.7505	0.7305	0.0227	0.6918–0.7556
Audio only	0.6450	0.6974	0.5870	0.6017	0.6224	0.6307	0.0386	0.5870–0.6974
Equal-weight fusion	0.7517	0.8108	0.7708	0.8090	0.7922	0.7869	0.0227	0.7517–0.8108
Fold validation-fitted fusion	0.7294	0.8070	0.7542	0.7932	0.7852	0.7738	0.0281	0.7294–0.8070

The session-level values preserve the aggregate ordering without implying uniform difficulty. Three-turn context exceeded utterance-only text in all five sessions, and equal-weight fusion achieved the highest macro F1 in every session. Audio varied most across sessions (SD 0.0386; range 0.5870–0.6974), whereas contextual text and equal fusion each had an SD of approximately 0.0227. These are descriptive results for five actor dyads, not estimates of population-level speaker variability; the dialogue-cluster bootstrap intervals remain the inferential comparison reported for RQ1 and RQ2.

Figure 5.1 places the classification results and calibration trade-off on common visual scales. The utterance-only condition is omitted from the calibration panel because no canonical ECE is reported for that configuration.

The row-normalised confusion matrices in Figure 5.2 expose the class-specific complementarity behind the aggregate result. Audio recognised anger more often than contextual text (78.1% versus 69.6% recall), whereas contextual text was substantially stronger for happiness and neutral. Equal fusion raised recall to 85.1% for anger, 81.1% for happiness, 66.6% for neutral, and 85.8% for sadness. Fold-specific validation fitting retained a more balanced pattern while producing probabilities suitable for operational confidence.

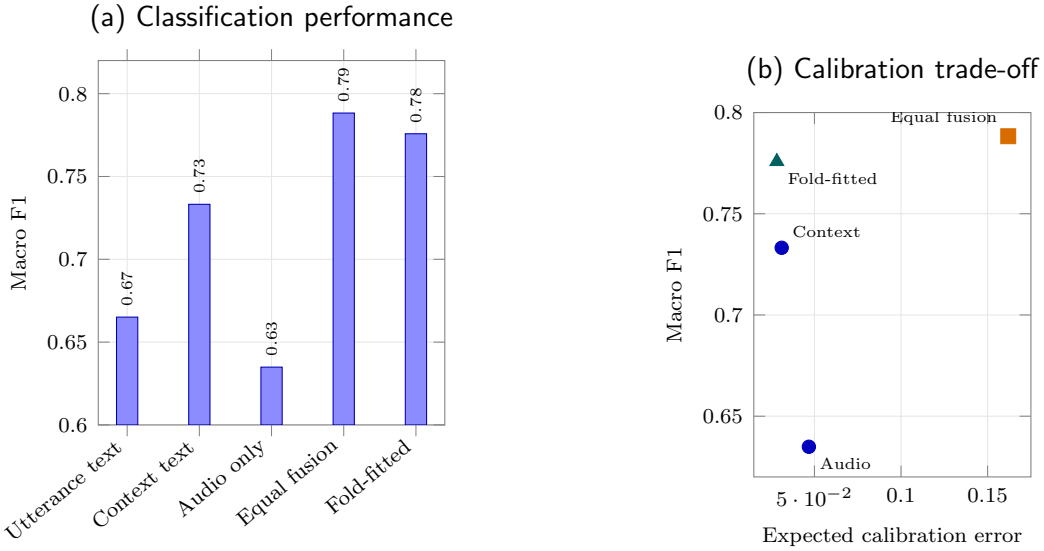


Figure 5.1: Aggregate four-class results from independently evaluated fold configurations. Panel (a) compares macro F1, while Panel (b) shows why the highest-classification equal-weight fusion is not the preferred configuration when confidence is consumed operationally.



Figure 5.2: Pooled row-normalised confusion matrices over the five held-out IEMOCAP sessions ($n = 5{,}531$). Rows are true labels and columns are predicted labels; cell values are percentages of the corresponding true-label row. All panels use the same 0–100% colour scale. Every fitted parameter represented here was estimated without the corresponding test session.

Figure 5.3 shows the corresponding bin-level calibration behaviour. Equal fusion lies systematically above the ideal diagonal through its populated mid-confidence region, indicating under-confidence despite its superior classification performance. Fold-specific validation fitting moves the fusion curve much closer to the diagonal. The isolated low-confidence points contain very few observations and should not be interpreted as stable local estimates.

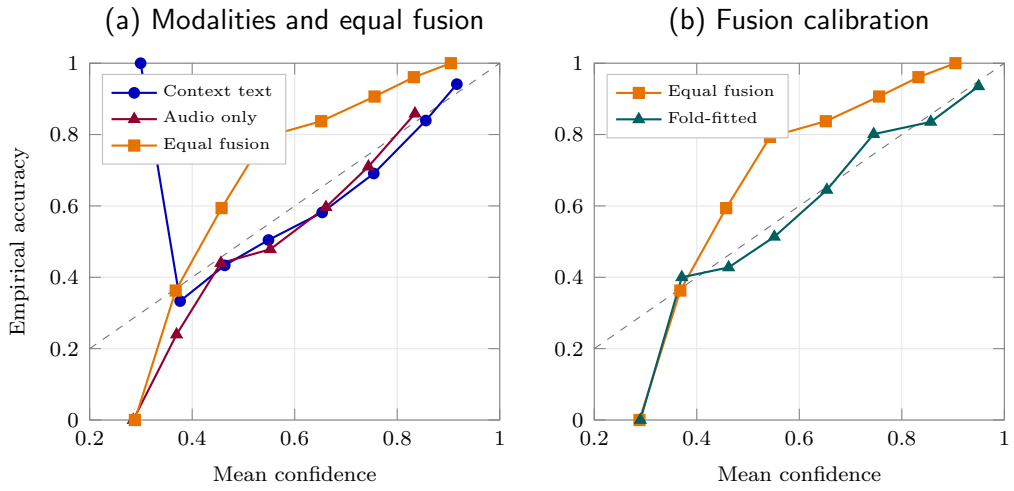


Figure 5.3: Reliability curves from pooled held-out predictions. Each point represents a non-empty equal-width bin of maximum class probability; the dashed diagonal denotes perfect calibration. Panel (a) compares the unimodal systems with predeclared equal fusion, and Panel (b) shows the effect of validation-derived fusion calibration. ECE values in Table 5.8 use the prespecified 15-bin calculation, while the curves use ten bins for legibility.

Operational full-data packaging analysis

The application requires one calibration chain for the final text and audio models rather than five fold-specific chains. For that purpose, the complete global chain fitted text and audio temperatures and then fusion parameters on 5,531 pooled validation-role predictions. It selected text weight 0.59, audio weight 0.41, and final temperature 0.5989.

Table 5.11: Operational global-chain summary; these descriptive values are not an independent held-out performance estimate.

Quantity	**Value**
Accuracy on pooled test-role predictions	0.7736
Macro F1 on pooled test-role predictions	0.7776
Weighted F1 on pooled test-role predictions	0.7728
ECE on pooled test-role predictions	0.0242
Negative log-likelihood	0.6075
Multiclass Brier score	0.3280

These values describe the packaged transformation but are label-reused. Across the rotating folds, the global fit used a validation-role prediction and label for every one of the 5,531 utterance identities; the summary then used a test-role prediction and the same label for every one of those identities, generated by a different trained model. Changing the model that produced the prediction does not make the label independent of calibration fitting. The global row is therefore excluded from Table 5.8 and from the held-out diagnostic figures. The equal-weight result remains the classification headline, while the fold-specific validation-fitted result supplies the held-out calibration evidence.

5.5.5 Comparison with Published IEMOCAP Results

Published IEMOCAP scores are comparable only when the target labels, example selection, speaker separation, test folds, modalities, and metric definitions are aligned. Li et al. provide the closest published reference for the present experiment: their study uses the same four labels, merges excitement with happiness, reports 5,531 utterances, and evaluates audio, text, and audio–text systems by leave-one-session-out five-fold cross-validation [LWC+22]. In their terminology, weighted accuracy (WA) is overall accuracy and unweighted accuracy (UA) is the unweighted mean of the four class recalls. For Table 5.12, the AffectLab UA values were therefore computed from the pooled confusion matrices; UA is not macro F1 and should not be substituted for the dissertation's primary metric.

Table 5.12: Descriptive comparison with protocol-compatible four-class, session-held-out IEMOCAP results reported by Li et al. [LWC+22]. Values are percentages.

System	Modality	Accuracy/WA	UA
AffectLab wav2vec 2.0 baseline (pooled held-out predictions)	Audio	62.79	65.27
Li et al., fine-tuned WavLM+Transformer encoder (five-fold mean)	Audio	67.99	68.24
AffectLab three-turn causal DeBERTa-v3 (pooled held-out predictions)	Text	73.04	73.32
Li et al., fine-tuned BERT+fully connected classifier (five-fold mean)	Text	77.67	78.76
AffectLab predeclared equal-weight late fusion (pooled held-out predictions)	Audio+text	78.32	79.64
Li et al., fine-tuned multimodal Transformer (five-fold mean)	Audio+text	80.36	81.70

The shared mapping and session-held-out test design make this a substantially closer comparison than results obtained from random utterance splits. The same broad pattern appears in both studies: text exceeds audio, and multimodal fusion gives the strongest classification result. AffectLab's equal-weight fusion falls below the published multimodal Transformer on both displayed measures, while remaining in the same performance range. This is benchmark placement, not a controlled head-to-head ranking. Li et al. average the five fold scores, whereas the AffectLab entries concatenate the five non-overlapping test sessions before computing the metrics; exact percentage-point differences between rows should therefore not be interpreted as like-for-like effect estimates. Their development set is a randomly selected speaker from the four non-test sessions, while AffectLab reserves a complete session for validation and trains on the other three. Their text context also includes four future as well as four preceding utterances, whereas AffectLab uses only the previous three turns so that inference is causal. Architectures, audio truncation, optimisation, and random variation differ as well; consequently, the table does not isolate the effect of any one modelling choice and no significance claim is made between studies.

Several commonly cited IEMOCAP results are deliberately excluded. The 84.45% WA and 84.39% UA reported by Li et al. for Session 5 use a single test session rather than the full five-fold evaluation, so direct numerical comparison with the pooled five-session AffectLab result is invalid [LWC+22]. DialogueGCN, COSMIC, and CoMPM retain six separate IEMOCAP labels, including excitement and frustration, and use a fixed training/development/test partition rather than the four-class rotating-session task [GMP+19, GMG+20, LL22]. Because both their

label mapping and their split differ, direct numerical comparison with the four-class results in this dissertation is invalid. The same restriction applies to random utterance-level cross-validation, which can place speakers from one session on both sides of a split. These exclusions prevent superficially similar IEMOCAP percentages from being presented as though they measured the same prediction problem.

5.5.6 Answer to RQ2

RQ2 is also answered positively within IEMOCAP. Equal-weight late fusion improved macro F1 by 0.0551 over contextual text and by 0.1534 over audio alone, showing complementary information in the textual and acoustic branches. The paired interval against contextual text excluded zero.

The answer includes a material trade-off: the strongest classifier was poorly calibrated, whereas fold-specific validation fitting sacrificed approximately one percentage point of macro F1 while reducing ECE substantially on held-out sessions. AffectLab uses the separately reported global chain as an operational packaging choice and retains conservative presentation rules; its label-reused summary is not additional held-out evidence. Calibration under IEMOCAP does not guarantee calibration after distribution shift [OFR+19].

5.6 Application Verification and Readiness

Table 5.13 provides a compact trace from the requirements in Chapter 3 to their implementation and verification evidence. Requirements are grouped only where they share the same implementation boundary, verification method, outcome, and principal limitation. Here, "passed" means conformance under the cited developer-led checks, not universal validation beyond the tested configuration. For application-readiness claims, that configuration is the offline deterministic core: the default suite uses the memory repository, while the full-stack smoke journey adds a compiled frontend and disposable MongoDB instance; both disable live providers and trained inference. It is a verification configuration, not a validated production environment.

5.6.1 Deterministic Dialogue Cases

The versioned engineering evaluator contains five authored trajectories for enhanced workload, boundary, and relationship practice. Each specifies the user turns and expected sequence of controller stages. All five enhanced trajectories matched their expected sequence, as shown in Table 5.14. The fixed controller is included only to expose the change in policy granularity; it stops according to its own earlier completion conditions and is not treated as a human-performance baseline.

These cases show that the enhanced controller requires intermediate evidence rather than completing as soon as one generic success phrase is detected. They do not show that its dialogue is more persuasive, realistic, or educational for a participant. Those properties require human evaluation.

Table 5.13: Compact requirement-to-evidence traceability matrix.

Requirement IDs	**Implementation location**	**Verification method**	**Outcome**	**Principal limitation**
FR1, FR26; NFR5, NFR13	Authentication, persistence, isolation, and deletion (Sections 4.3, 4.3, and 4.6)	API, security, ownership, stale-write, deletion, and MongoDB recovery tests	Passed	No independent penetration test or deployed backup exercise
FR2, FR3, FR25; NFR3	Onboarding, preferences, takeaways, and action cards (Sections 4.5 and 4.5)	API and browser onboarding, preference, reflection, and recovery journeys	Passed	No independent usability assessment of these controls
FR4–FR8, FR15, FR19	Preparation, scenario configuration, lifecycle, and dialogue policies (Sections 4.4, 4.4, and 4.4)	Policy/unit, API, desktop/mobile browser, authored-trajectory, and smoke tests	Passed	Three standard scenarios and controlled inputs
FR9, FR11–FR13; NFR2, NFR4	Voice, causal-context inference, fusion, and fallback paths (Sections 4.2, 4.5, and 4.5)	Context, multimodal, transcription, transient-audio, missing-service, and browser tests	Passed in tested configuration	Limited live-device, provider, and private-model execution
FR10; NFR1	Crisis interruption and safety precedence (Section 4.3)	Crisis paraphrase, false-positive, precedence, API, and conversation tests	Passed	A bounded lexical detector cannot cover every expression
FR29	Cognitive heuristics and strategy selection (Subsection 4.3)	Authored unit and service cases for pattern, intent, clause extraction, pushback, strategy, and safety precedence	Passed as rule conformance only	No construct, predictive, clinical, comparative-helpfulness, or harm evaluation
FR14; NFR14	Bounded affect pacing and uncertainty presentation (Sections 4.4 and 4.5)	Seven authored policy cases plus preference, disagreement, failure, and replay tests	Passed	Controlled probabilities; no evidence of user benefit
FR16–FR18, FR20–FR21; NFR6, NFR8	Evidence, response realisation, fallback, and feedback (Sections 4.4, 4.4, and 4.5)	Determinism, evidence-linking, constrained-generation, fallback, feedback, and concurrency tests	Passed	Designed cases do not exhaust language variation
FR22–FR24	Replay, retry, branching, and comparison (Sections 4.5 and 4.5)	Backend, browser, stale-state, restart-persistence, and full-stack smoke tests	Passed	Synthetic histories rather than natural long-term use
FR27–FR28; NFR7	Consent, study progress, pseudonymous export, and versioning (Section 4.5 and Subsection 4.2)	Protocol, eligibility, withdrawal, export-exclusion, dataset-freeze, provenance, and smoke tests	Passed	No independent privacy or reproduction audit
NFR9	Health, bounded failures, recovery, and local responsiveness (Subsections 4.6 and 4.6)	Recovery journeys, health checks, full-stack restart, and performance regression artefact	Passed locally	No networked concurrent-load or provider-latency test
NFR10	Accessible and responsive interface (Subsections 4.5 and 4.5)	axe-core scans, keyboard assertions, and desktop/mobile browser runs	Passed tested views	No evaluation with users with disabilities
NFR11–NFR12	Modular and testable architecture (Sections 4.3 and 4.6)	Interface inspection, provider substitution, and automated suites across component boundaries	Shown	Maintainability was not measured longitudinally

Table 5.14: Authored enhanced-dialogue trajectory results.

Case	Expected and observed enhanced stages	Final state
Workload agreement	constraints → agree → resolved	Completed: success
Workload non-confirmation	constraints → agree → agree	Active; no false agreement
Boundary maintained	pressure → resolved	Completed: success
Relationship next step	perspective → agree → resolved	Completed: success
Relationship unresolved	perspective → unresolved	Completed: unresolved

5.6.2 Cognitive-Heuristic and Support-Routing Cases

The cognitive and strategy tests are small, implementation-authored examples rather than a labelled evaluation dataset. Table 5.15 states exactly what they establish. The combined sentence deliberately activates several rules at once; the test verifies that the documented precedence and extraction code are stable, not that a human assessor would assign the same psychological labels.

Table 5.15: Implemented cognitive-heuristic and support-routing checks.

Authored input or condition	Verified implementation result	What the check does not establish
"They will definitely think I am a failure because I missed one deadline. What should I do?"	A cognitive-pattern rule matches; practical-help wins the intent precedence; the clause after *because* is copied as possible context	That the speaker has a cognitive distortion, that the copied clause is the true cause, or that practical advice is wanted
"But that won't work. This is pointless."	The explicit pushback vocabulary produces a resistance value above the selector threshold	Psychological resistance, its degree, persistence, or cause
"How do I handle this?" with the default lexical affect state	Practical-help routing makes practical suggestion the highest-scoring strategy	That the suggestion is helpful, well timed, or superior to clarification or reflection
Crisis flag true with high arousal	Safety escalation wins over all ordinary strategy scores	Sensitivity or specificity of crisis detection, or adequacy of the safety response
Two authored emotional-support messages	The analyser, selector, and offline template compose a non-generic emotion-specific response	Accuracy of the affect or intent label, user preference, therapeutic value, or absence of harm

These checks cover selected positive paths and one precedence condition. They do not systematically test all ten pattern families, all combinations and tie cases, negated and quoted pattern language, malformed causal clauses, demographic or dialectal variation, or user correction of a wrong inference. The browser suite verifies rendering of deterministic decision data but does not independently validate the semantics of the labels. Consequently, the appropriate result is "conforms to the tested rules", not an accuracy percentage. The subsystem did not contribute an outcome measure to the pilot, and no claim is made that its labels or selected strategies improved the participant experience.

5.6.3 Evidence Extraction and Affect-Pacing Cases

Twelve human-authored feature judgements were compared with the deterministic extractor. The fixture produced five true positives, four true negatives, one false positive, and two false negatives. Reported speech containing "I need" was incorrectly attributed to the current speaker; an indirect refusal and the synonym "shift" were missed. With only twelve designed cases, these counts are diagnostic examples rather than a held-out accuracy estimate. Importantly, the errors expose the intended limitation of a bounded lexical policy and justify replaying the complete source utterance.

Seven controlled affect cases exercised matching confident estimates, modality disagreement, low confidence, missing audio, an unavailable model, disabled adaptation, and an explicit user choice. All seven produced the declared action. Matching high-confidence sadness allowed a brief acknowledgement; disagreement and low confidence offered a pacing choice; unavailable evidence and disabled adaptation preserved the baseline; and the explicit gentler preference took precedence. Separate tests verify that safety, completion, ineligible sessions, and required study tasks override adaptation.

The deterministic evaluator recorded median in-process service latency of approximately 0.64 ms and a lower-order-statistic 95th-percentile value of approximately 3.46 ms across 19 calls. These figures exclude HTTP, MongoDB, provider generation, and trained inference and are not a load test or production latency claim.

5.6.4 Automated and Full-Stack Verification Results

The final application verification was repeated on 19 October 2026 against commit `2e275767f412ea858d54ac2c507a189815581785`. The worktree was clean when testing began; the subsequent changes in this subsection alter reporting only. The machine ran Windows 11 build 26300 on an AMD Ryzen 5 3500U with eight logical processors and 5.9 GiB of physical memory. The relevant toolchain was Python 3.14.3, pytest 8.4.2, Ruff 0.14.10, Node.js 24.14.1, npm 11.12.1, Playwright 1.63.0, axe-core 4.13.0, and bundled Chromium 153.0.8010.12. No GPU-dependent path was measured. The separately saved full-stack manifest records MongoDB 8.0.1.

The exact final verification commands, issued from the repository root unless the listing changes directory, were:

```
.\.venv\Scripts\python.exe -m pytest backend\tests -rs
.\.venv\Scripts\ruff.exe check backend scripts
Set-Location frontend
npm.cmd run build
npm.cmd run lint
$vite = Start-Process -FilePath node.exe -ArgumentList '.\node_modules\vite\bin\vite.js','--host
    ','127.0.0.1','--port','4173' -WorkingDirectory (Get-Location) -WindowStyle Hidden -PassThru
npx.cmd playwright test replay.spec.ts branching.spec.ts preparation.spec.ts practice.spec.ts
    workspace-recovery.spec.ts --workers=2
Stop-Process -Id $vite.Id
```

Starting Vite separately avoided a Windows teardown wait observed when Playwright owned the web-server process; it did not change the tests or frontend source. The production TypeScript/Vite build, ESLint, and Ruff checks exited successfully. Pytest collected 258 backend cases: 245 passed and 13 were skipped in 88.07

seconds. The focused browser command expanded 38 logical cases across desktop Chromium and the Pixel 7 mobile profile, so 76 browser cases passed in 3.3 minutes. These totals count parameterisations and browser projects as executed test cases; skipped backend cases are not counted as passes.

All 13 skips were the following real-MongoDB node IDs from `backend/tests/test_mongo_integration.py`; each had the explicit reason `Set TEST_MONGODB_URI to run real MongoDB tests`:

```
test_prepared_brief_and_card_survive_restart_without_changing_feedback
test_branch_restart_concurrent_creation_and_parent_deletion
test_data_persists_when_repository_client_restarts
test_ttl_index_and_logical_expiry_do_not_remove_research_record
test_account_deletion_cascades_across_personal_collections
test_concurrent_session_updates_reject_a_lost_update
test_backup_can_be_restored_and_verified
test_password_reset_rejection_preserves_link_and_concurrent_success_is_single_use
test_enrolled_rehearsal_round_trip_and_restart
test_enhanced_workload_state_survives_restart
test_profiled_dialogue_reload_continue_and_rewind[boundary-I cannot take this on.-continuation0-
    boundary_held]
test_profiled_dialogue_reload_continue_and_rewind[relationship-When we spend evenings apart, I feel
     lonely and need time together.-continuation1-next_step]
test_exchange_bound_predictions_survive_restart_without_audio
```

The skips therefore represent one declared environmental boundary rather than failures distributed across unrelated functionality. Real-database behaviour is instead evidenced by the separately recorded smoke journeys and the earlier opt-in MongoDB run; the 19 October default-suite total does not silently treat those 13 cases as successful.

Accessibility assertions used axe-core 4.13.0 through `@axe-core/playwright` 4.13.0. The selected tags were `wcag2a`, `wcag2aa`, `wcag21a`, and `wcag21aa`, corresponding to automated rules mapped to WCAG 2.0 and WCAG 2.1 Levels A and AA. Ten distinct interface states were scanned in both browser projects, giving 20 scan executions: the restored reflection after voice submission; paused enhanced-workload progress; paused boundary-pressure progress; reviewed voice text with pacing; stored prediction provenance; editable preparation brief; saved action card; evidence-linked replay; uncertain multimodal replay; and branch comparison. These states covered `/app`, preparation and active-session variants of `/practice`, Replay, Compare, and Action Card views. Each assertion required an empty axe violation array, and all 20 returned zero violations. This result is limited to the rules, DOM states, fixtures, browsers, and viewports exercised; it is not a WCAG conformance certification and does not replace screen-reader, assistive-technology, or participatory testing.

The saved full-stack smoke manifest was recorded on 4 October 2026 and incorporated with its source-hashed inputs at commit `00a7f54aecefec1d8bb6f53235ddc7ab8f4f0773`. It was generated from the repository root by the exact command `.\.venv\Scripts\python.exescripts\engineering_smoke.py`. The manifest reports two passing browser journeys against a compiled frontend, actual FastAPI process, and disposable MongoDB 8.0.1 database. The journeys included account access, pre-consent exclusion, enrolment, a required task and questionnaire, researcher export, enhanced dialogue, network-failure recovery, replay, branching, comparison, and action-card persistence. The backend was stopped and relaunched twice while MongoDB remained active, and persisted state was verified after each relevant restart. The recorded total run time was 95.72 seconds; this is demonstration timing, not a

capacity benchmark.

Together, these complementary checks support the requirement outcomes summarised in Table 5.13; their principal scope limitations are reported in the same table.

5.6.5 Performance and Responsive-Interface Checks

A reproducible local benchmark exercised the HTTP boundary of the deterministic core using FastAPI's in-process transport and the memory repository. It included routing, bearer-token authentication, request validation, application services, and response serialisation. The evidence was recorded on 11 October 2026, incorporated with its source-hashed inputs at commit `230073c6026f73b24c9162787f3ff493b7ed8a81`, and generated from the repository root by `.\.venv\Scripts\python.exescripts\evaluate_performance.py`. After warm-up, the benchmark collected 200 samples for read-only endpoints and 100 samples for each state-changing workflow operation. Table 5.16 reports the results from the recorded Windows 11 and Python 3.14.3 run.

Table 5.16: Local single-client application latency results.

Operation	**Samples**	**Median (ms)**	**P95 (ms)**	**P95 budget (ms)**
Liveness endpoint	200	1.332	1.866	25
Authenticated scenario catalogue	200	2.942	3.922	50
Session creation	100	2.988	3.961	50
Role-play start	100	3.396	4.437	75
Deterministic conversation turn	100	4.659	5.836	100

Each sample was a sequential, single-client request timed with Python's `time.perf_counter` around FastAPI `TestClient.request`; response status validation remained inside the measured operation. One liveness and one scenario-catalogue request warmed routing, schema, and service paths before collection. The median is Python's conventional sample median. P95 uses the nearest-rank rule: after ascending sort, the observation at one-indexed rank $\lceil 0.95n \rceil$ is selected (rank 190 of 200 or 95 of 100). A check passes when this P95 is less than or equal to its budget. The five budgets in Table 5.16 were fixed in the benchmark script before execution as deliberately generous regression tripwires; they are not production service-level objectives derived from users or deployed traffic.

The benchmark is a fast regression check for local application overhead, not a production-capacity test. It excludes network latency, MongoDB, external generation and transcription providers, trained multimodal inference, and concurrent clients. Those variables depend on the deployed environment and require a separate staged load test. Within its declared scope, however, the result shows that the locally controlled text-first path adds only a few milliseconds per request and that optional services are not on the critical path when deterministic fallbacks are used. All five measurements passed their P95 regression budgets. The JSON evidence artefact records the environment, sample counts, budgets, source hashes, individual scope exclusions, and summary statistics.

Perceived responsiveness also depends on the browser rather than API latency alone. The frontend therefore runs as a compiled production build in both desktop Chromium and a Pixel 7 mobile profile. The browser suite verifies responsive layouts, keyboard interaction, focus behaviour, live status messages, preservation of unsent input after failure, and immediate recovery controls. These checks support the claim that the interface is responsive across the tested viewports; they do not replace field measurements of Core Web Vitals on the eventual hosting infrastructure.

5.6.6 Answer to RQ3

RQ3 asks how uncertain multimodal predictions can be integrated while keeping behaviour explainable, controllable, safe, and failure-tolerant. The implementation and engineering evidence provide a constructive answer: the classifier output is validated and stored as evidence; a deterministic policy maps confidence, agreement, availability, eligibility, and user preference to a small action set; the dialogue controller remains authoritative; and each action and fallback reason is persisted for replay.

The authored cases and automated tests show that low confidence and disagreement produce a user choice, missing inference preserves the baseline, explicit preference takes precedence, and safety bypasses model adaptation. Version-bound requests and optimistic persistence attach a prediction to its exact exchange. This establishes specified system behaviour under the tested conditions. It does not establish that the emotion label itself is correct for a real participant or that pacing improves their outcome.

5.6.7 Answer to RQ4: Integrated Application Quality

RQ4 is answered through the combined evidence traced in Table 5.13. The automated, browser, full-stack, recovery, and authored-policy checks passed for the offline deterministic core defined above, while its locally controlled text-first operations remained below 7 ms at the 95th percentile. These results establish implementation completeness for the tested paths and make the system suitable for controlled deployment only after environment-specific validation.

That validation must exercise the intended MongoDB deployment, trained inference and external providers, real network transit, expected concurrency, backup and recovery, observability, and adversarial and independent security testing. It should establish service-level objectives and close the private model-file provenance gaps before deployment. The present evidence therefore does not claim production latency, internet-scale capacity, or security assurance.

5.6.8 Supplementary User-Pilot Results

Scope of the pilot result. The pilot did not evaluate multimodal affect adaptation. Every required task was ineligible for multimodal inference and affect-aware pacing, so participants completed the baseline text-based rehearsal with deterministic dialogue and feedback. Accordingly, the ratings below describe that baseline

experience only: they cannot establish whether emotion-aware adaptation is acceptable, useful, or beneficial. Evidence concerning the classifier and the implemented adaptation policy instead comes from the IEMOCAP experiments in Section 5.5 and the authored, automated, and full-stack tests reported above; those tests establish benchmark performance and policy conformance within their stated scopes, not human benefit.

Within that boundary, the supplementary single-arm pilot examined completion of the baseline rehearsal workflow and collected brief usability ratings. The analysis uses the AffectLab feasibility protocol v1.1 export, with the first attempt at each required intermediate-difficulty scenario selected for the protocol outcomes. The analytic export includes 15 participants and 46 rehearsal sessions, of which 42 were first required attempts and four were additional or retry sessions. Of the required attempts, 39 reached task completion (14 workload, 13 boundary, and 12 relationship). Twelve participants completed all three required scenarios and submitted the corresponding post-rehearsal questionnaires, giving a protocol-completion rate of 80.0% (12/15; two-sided 95% Wilson interval 54.8%–93.0%). The point estimate met the protocol's 80% feasibility target, although the interval remained wide at this sample size.

Table 5.17 summarises the available study-specific ratings. Feedback usefulness and scenario realism for the non-adaptive required tasks had means of 5.33 and 4.85, respectively, on seven-point scales. These scores concern the deterministic feedback and scenario experience, not multimodal inference or affect-aware pacing. Among 38 observations with both confidence ratings available, representing 14 participants, the mean within-task confidence change was +0.97 points (median +1; participant-cluster bootstrap 95% interval +0.67 to +1.26). The mean paired changes were +0.85 for workload ($n = 13$), +0.85 for boundary ($n = 13$), and +1.25 for relationship ($n = 12$). These within-task differences reflect immediate self-report, not independently verified communication improvement or an effect attributable to the application.

Table 5.17: Supplementary pilot questionnaire results from available first required attempts. Ratings use seven-point scales; paired change is post-confidence minus pre-confidence.

Measure	n	**Mean**	**SD**	**Median**	**Q1–Q3**
Pre-rehearsal confidence	40	3.80	1.04	4.0	3.0–4.0
Pre-rehearsal anxiety	40	4.58	1.28	5.0	4.0–5.25
Post-rehearsal confidence	40	4.75	1.37	4.5	4.0–6.0
Scenario realism	40	4.85	0.98	5.0	4.0–5.25
Feedback usefulness	40	5.33	0.92	5.0	5.0–6.0
Paired confidence change	38	+0.97	0.97	+1.0	0.25–1.75

The deterministic feedback also produced scenario-specific lexical indicators on the 39 completed required tasks. Mean scores included 42.3% for clear requests in workload rehearsal ($n = 14$), 59.0% for clear refusal or maintained boundaries ($n = 13$), and 44.4% for use of an I-statement in relationship rehearsal ($n = 12$). The configured absence-of-blame indicators reached 100% in workload and relationship tasks, and the no-repeated-apology indicator reached 100% in boundary tasks. These are averages of rule-defined scores rather than percentages of participants demonstrating validated skills; the indicators were not independently annotated or

established as measures of real-world communication ability. They provide limited behavioural context to the stronger engineering evidence for the feedback implementation.

Two required attempts lacked a post-questionnaire, and only 38 of the 42 required attempts contained a valid pair of confidence ratings. Consequently, questionnaire summaries use available observations and report their denominators rather than imputing responses. The pilot offers preliminary evidence of workflow uptake and perceived usefulness, but the small convenience sample and uncontrolled design preclude causal, clinical, or generalisable effectiveness claims. It is therefore treated as a supplement to the model and system-verification results, not as the principal basis for deployment readiness.

5.7 Threats to Validity and Limitations

5.7.1 Model and Statistical Validity

The five test folds cover all IEMOCAP sessions, but the corpus contains only ten actors. Fold results are therefore dependent on a small set of speakers, acted or improvised tasks, recording conditions, and annotation practices. The dialogue bootstrap reflects the 151 observed conversations but cannot compensate for missing demographic, cultural, linguistic, and environmental diversity. Only one primary training seed was used after the pipeline check, so optimisation variance is not fully characterised.

Repeated GPU execution can produce small numerical variation. The reported paired comparisons use the latest internally consistent private prediction files, each checked against its neighbouring metrics record. This preserves internal pairing but illustrates why commit, configuration, and file hashes are necessary. The final full-data packages cannot be evaluated on their training rows, and their file-level provenance remains incomplete until the private directories are downloaded and hashed beside the application.

ECE is bin-dependent, and calibration can deteriorate under shift. The global deployment chain is fitted from pooled validation roles across the corpus; it is useful for choosing one full-data configuration but is not an independent external calibration study. The 0.55 and 0.75 presentation bands are operational thresholds rather than validated psychological boundaries.

5.7.2 External and Construct Validity

IEMOCAP labels describe annotated expressive categories, not direct measurements of internal emotion. Mapping excitement to happiness and fear to exploratory anxiety further simplifies the constructs. Model performance must not be interpreted as diagnosis, emotional truth, or equitable performance across accents, languages, devices, ages, disabilities, or cultures.

The role-play features are likewise operational constructs. A phrase detector can identify authored patterns such as a clear refusal or practical request, but it does not measure general communication ability or transfer to a real conversation. The diagnostic fixture already demonstrates missed synonyms, indirect language,

and reported-speech error. Deterministic feedback is more auditable than an unconstrained judgement, but it remains limited by these definitions.

The rule-based cognitive fields have an even narrower evidential basis. Familiar labels such as cognitive distortion, intent, cause, readiness, and resistance may suggest assessment of an internal state, but the implementation observes only the current message and assigns fixed values when short patterns occur. No clinician, validated questionnaire, independent annotator, held-out corpus, or participant self-report was used as a reference standard. The tests reported in Subsection 5.6.2 establish software conformance, not accuracy, construct validity, or clinical validity. Because the pilot neither isolated this subsystem nor measured its helpfulness or harms, the selected support strategy cannot be claimed to be optimal or beneficial. False and overly authoritative labels remain a risk, especially for negation, quotation, indirect language, cultural variation, and messages containing several possible requests.

The engineering evaluation largely uses deterministic templates and test accounts. It does not evaluate live provider variability, private-model latency under concurrent participant use, real microphone noise, or long-duration deployment. The successful smoke journey establishes integration and restart persistence only for its tested path.

5.7.3 Supplementary User-Evidence Limitations

The supplementary pilot was single-arm and included only 15 participants in the analytic export, limiting precision and generalisability. More fundamentally for the present dissertation, required tasks bypassed multimodal inference and affect-aware pacing. The pilot therefore provides no direct user evidence about whether the emotion-aware adaptation was acceptable, accurate, intrusive, helpful, or harmful; its realism and usefulness scores apply only to the baseline rehearsal and deterministic feedback that participants encountered. The wide confidence interval around protocol completion illustrates additional uncertainty. Missing ratings, incomplete rehearsals, fixed scenario order, repeated exposure, self-selection, and demand characteristics may influence the descriptive findings. The study-specific ratings are not validated clinical scales, and the deterministic lexical scores were not validated against independent assessments of communication quality. Without a control condition or follow-up assessment, the observed immediate confidence change cannot establish causation, sustained skill development, or transfer to real conversations. These limitations are why the pilot contributes contextual usability evidence rather than the principal validation of the application or evidence for affect-aware adaptation.

5.8 Chapter Summary

The evaluation provides completed answers to the two machine-learning questions. Three-turn causal context improved the four-class text model from 0.6651 to 0.7332 macro F1, with a paired dialogue-bootstrap interval excluding zero. Equal-weight text–audio fusion further improved macro F1 to 0.7883, demonstrating complementary benchmark information. However, its ECE of 0.1619 exposed a classification–calibration trade-off. Validation-derived calibration reduced ECE substantially at a

modest classification cost, motivating the global text-dominant configuration used for final packaging.

The application evaluation answers RQ3 and RQ4 at the level of specified, reproducible system behaviour. All five authored enhanced dialogue trajectories and all seven affect-pacing decisions matched expectations. Diagnostic evidence cases exposed one false positive and two false negatives rather than concealing lexical limitations. A backend run with 245 passes and 13 explicitly identified environmental skips, 76 passing desktop/mobile browser cases, 20 zero-violation automated accessibility scans, production frontend checks, two full-stack journeys, persistence across backend restarts, and local HTTP timings verify complementary functional and non-functional boundaries. The deterministic conversation path recorded a median of 4.659 ms and a 95th percentile of 5.836 ms in the scoped local benchmark.

The small pilot remains supporting evidence for the non-adaptive workflow: 12 of 15 participants completed the three required rehearsals and questionnaires, while mean scenario-realism and feedback-usefulness ratings were 4.85 and 5.33 out of seven. Because those required rehearsals bypassed multimodal inference and affect-aware pacing, these observations say nothing about the acceptability or benefit of emotion-aware adaptation. The evaluation therefore rests principally on the separate model benchmark and the authored, automated, browser, persistence, recovery, full-stack, accessibility, and performance tests. Overall, AffectLab is a fully implemented communication-practice system whose model behaviour, core policies, interfaces, persistence, accessibility, recovery, and local responsiveness have been tested extensively by the developer and through automation, with human evaluation of adaptation and deployment-scale performance appropriately reserved for future controlled and staging studies.

Chapter 6

Conclusions and Future Work

6.1 Chapter Overview

This dissertation investigated how multimodal emotion recognition can be incorporated into a helpful application for rehearsing difficult interpersonal conversations without allowing uncertain predictions to control consequential decisions. The work combined speaker-independent emotion-recognition experiments, an uncertainty-aware interaction policy, structured role-play scenarios, evidence-grounded feedback, a production-oriented full-stack application, and a layered developer-led verification strategy. AffectLab demonstrates one way of using affective information as bounded conversational evidence rather than as a declaration of a user's internal state.

This final chapter synthesises the findings in relation to the four research questions, identifies the principal contributions, and restates the boundaries of the conclusions. It then proposes a prioritised programme of future work spanning deployment-scale performance, operational governance, model generalisation, policy comparison, feedback validation, accessibility, and longer-term product research. The chapter closes by considering what the implemented and tested application establishes and what remains environment- or study-dependent.

6.2 Summary of the Dissertation

The dissertation began from two connected problems. First, emotion recognition in conversation is uncertain, sensitive to context, and vulnerable to distribution shift. Second, a role-play application needs predictable progression, evidence-grounded feedback, safety precedence, and user control even when machine-learning or generative services are unavailable. Treating these as separate problems would make it easy either to build a strong benchmark classifier with no responsible interaction design or to build an interactive application whose affective behaviour could not be evaluated independently.

AffectLab addresses this tension through separation of responsibilities. Contextual text and speech models estimate an affective probability distribution. A versioned adaptation policy considers confidence, agreement, availability, eligibility, and user preference before selecting a small pacing action. Deterministic scenario controllers remain authoritative for dialogue stage, completion, and unresolved out-

comes. Observable language features determine feedback metrics, while an optional language model is restricted to response realisation. Safety handling bypasses affect adaptation, and missing external services preserve the baseline text workflow.

The implementation supports three fixed scenarios: workload negotiation, boundary-setting, and expression of relationship needs. It also provides preparation, replay, rewind, branching, comparison, takeaways, and action cards so that a rehearsal can be inspected and revisited. Authentication, ownership checks, retention controls, transient voice handling, health checks, production configuration validation, containerised deployment, deterministic fallbacks, backup verification, and optional research-data controls make the application suitable for controlled production use and future study.

Evaluation was deliberately divided by claim. IEMOCAP experiments support classification and calibration findings on the benchmark. Authored trajectories, 245 passing backend tests with 13 identified environmental skips, production frontend checks, 76 passing desktop/mobile browser cases, 20 zero-violation automated accessibility scans, database-backed full-stack journeys, restart recovery, and local HTTP timings support application-quality claims. The scoped performance run recorded a 4.659 ms median and 5.836 ms 95th percentile for deterministic conversation turns. These model and engineering evaluations remain the principal evidence. Section 5.6.8 retains the small pilot only as supplementary feedback on the baseline rehearsal workflow: required tasks bypassed multimodal inference and affect-aware pacing, so the pilot did not evaluate the acceptability or benefit of emotion-aware adaptation.

6.3 Conclusions for the Research Questions

6.3.1 RQ1: The Contribution of Conversational Context

RQ1 asked whether preceding conversational turns improve text-based emotion recognition compared with independent utterance classification. Under the declared four-class, speaker-independent IEMOCAP protocol, the answer is yes. Adding the previous three causal dialogue turns increased accuracy from 0.6624 to 0.7304 and macro F1 from 0.6651 to 0.7332. The paired dialogue-bootstrap macro-F1 difference was +0.0681, with a 95% interval from 0.0504 to 0.0857. Accuracy and weighted-F1 intervals also excluded zero.

The result is strengthened by the controlled comparison: architecture, label set, data partitions, seed, and optimisation settings were held constant, leaving the input context as the intended experimental difference. The exploratory six-class comparison produced the same aggregate direction, although its anxiety category was based on only 40 mapped fear examples and cannot support a stable anxiety-recognition claim.

The conclusion remains task-specific. Context improved annotated emotion classification in IEMOCAP, but the experiment does not establish that the model recovers a person's private emotional state or that the same gain will transfer to spontaneous AffectLab conversations. Per-class results also show why the aggregate conclusion should not be generalised indiscriminately: the anger improvement was small and its interval included zero. The contribution of context is therefore

a benchmark classification finding rather than a claim of general emotional understanding.

6.3.2 RQ2: Multimodal Complementarity and Calibration

RQ2 asked whether contextual text and acoustic information are complementary and what trade-off exists between classification and confidence calibration. The audio-only model achieved macro F1 0.6349, below the contextual text result of 0.7332. Nevertheless, combining the two calibrated probability distributions with the predeclared equal-weight rule increased macro F1 to 0.7883 and accuracy to 0.7832. The macro-F1 improvement over contextual text was +0.0551, with a dialogue-bootstrap 95% interval from 0.0407 to 0.0713; all 10,000 bootstrap replicates favoured fusion. Acoustic information therefore contributed complementary benchmark information even though it was the weaker modality in isolation.

The strongest classifier was not the best calibrated configuration. Equal-weight fusion produced ECE 0.1619, compared with 0.0311 for contextual text. Fold-specific validation-fitted fusion reduced ECE to 0.0283 while retaining macro F1 0.7758 on held-out sessions. A separate global chain with text weight 0.59 was selected for operational packaging. Its pooled test-role summary (macro F1 0.7776 and ECE 0.0242) is not independent performance evidence: every utterance label used in that summary had already entered the global fit through the same utterance's validation-role prediction from another fold.

These results answer both parts of RQ2. Text and speech were complementary for classification, but probability combination sharpened confidence enough to require a separate calibration decision. Consequently, an application should not choose a fusion configuration solely because it maximises test-set F1. The distinction between classification performance and operational confidence is central when probabilities influence thresholds, user-facing uncertainty, or adaptive behaviour.

6.3.3 RQ3: Bounded Integration of Affective Evidence

RQ3 concerned the responsible integration of uncertain multimodal predictions into structured role-play. The project provides a constructive engineering answer. AffectLab stores the complete prediction and provenance, validates it against the current session and turn, and maps it through a deterministic policy to one of a restricted set of pacing actions. A fused or component confidence below the configured low-confidence boundary, or modality disagreement, offers the user a pacing choice; matching anger or sadness estimates for which all three confidences meet that boundary may permit a modest acknowledgement, including in the moderate presentation band; an explicit preference can preserve baseline behaviour; unavailable inference falls back without blocking the conversation; and safety interruption takes precedence over all affective adaptation.

This separation prevents the classifier from deciding whether a scenario succeeds, what feedback score is assigned, whether a participant is eligible, or whether a safety response is required. Optional language generation can phrase a selected action but cannot replace the recorded decision. Optimistic persistence, request identifiers, model versions, source fields, and fallback reasons make the resulting behaviour replayable and auditable.

The engineering evaluation supports this answer within the tested conditions. All five authored enhanced dialogue trajectories and all seven affect-pacing cases matched their expected outcomes. The full automated backend suite passed, and two integrated browser journeys completed against the disposable database while surviving actual backend restarts. These results establish policy conformance, recovery, ownership, persistence, and failure tolerance. They do not establish that an affect prediction was correct for a human user or that adaptation improves a user's rehearsal outcome.

6.3.4 RQ4: Integrated Application Quality and Readiness

RQ4 asked whether the completed application satisfies its declared functional and non-functional requirements under reproducible engineering evaluation. The answer is positive for the tested offline deterministic core. The default suite used the memory repository, and the full-stack smoke configuration added a compiled frontend and disposable MongoDB instance; neither configuration exercised live providers. The backend suite reported 245 passes and 13 identified real-MongoDB skips, the frontend passed strict production compilation and linting, 76 browser cases exercised desktop and mobile layouts, 20 embedded axe scans reported no violations, and two full-stack journeys verified real API serialisation, MongoDB persistence, network-failure recovery, branching, export, and state preservation across backend restarts. Authored cases additionally verified the deterministic dialogue and affect-pacing policies.

Responsiveness was tested separately from correctness. In a single-client, in-process HTTP benchmark, liveness, scenario retrieval, session creation, role-play start, and a deterministic conversation turn all remained below 7 ms at the 95th percentile. The benchmark excludes network transit, MongoDB, trained inference, providers, and concurrent users, so it supports a local-regression claim rather than a deployment-capacity claim. Taken together with fail-fast production configuration, health checks, ownership protection, optimistic concurrency, fallbacks, backup exercises, and containerisation, the evidence establishes an implementation-complete, deployment-oriented system. Suitability for a controlled deployment remains conditional on target-environment load, latency, security, observability, backup-and-recovery, and incident-response validation.

6.4 Principal Contributions

6.4.1 Empirical and Machine-Learning Contributions

The first contribution is a controlled, speaker-independent comparison of utterance-only and causal-context text emotion recognition. By pairing predictions at dialogue level and reporting aggregate, per-class, and exploratory six-class results, the study demonstrates a clear contextual benefit while preserving class-specific uncertainty.

The second contribution is a comparative text–audio pipeline that retains unimodal baselines and distinguishes predeclared fusion from validation-fitted operational configurations. Reporting calibration beside accuracy and F1 exposes a trade-off that would be hidden by classification metrics alone. The final pipeline

also records the temperatures, modality weights, label order, and provenance needed to reproduce the operational transformation from logits to probabilities.

6.4.2 System and Interaction-Design Contributions

The principal system contribution is a hybrid architecture in which trained models and generative services can influence presentation without becoming authoritative for dialogue outcomes, assessment, or safety. The uncertainty-aware affect policy turns confidence, agreement, availability, and preference into inspectable behaviour. Deterministic fallbacks preserve a useful interaction when optional services fail.

The three scenario controllers contribute explicit models of different interpersonal objectives. Workload negotiation seeks a practical prioritisation decision, boundary-setting recognises a respectfully maintained refusal without requiring compliance, and relationship practice seeks a specific need or next step while allowing disagreement to remain unresolved. This avoids reducing every role-play to one generic success condition.

Evidence-grounded feedback and reflective tools form a further applied contribution. Metrics point to specific conversational turns rather than issuing an unexplained global judgement. Replay, rewind, branching, and comparison support counterfactual practice from the same context. Together, these features make a conversation an inspectable practice artefact instead of a transient exchange ending in an opaque score.

6.4.3 Application-Verification and Infrastructure Contributions

The dissertation contributes a layered verification framework that keeps model performance, application conformance, performance scope, and supporting user feedback separate. This prevents benchmark accuracy from being interpreted as product reliability and prevents a small pilot from overshadowing the much broader technical verification. The framework combines deterministic fixtures, automated unit and API suites, real persistence checks, desktop/mobile browser and accessibility tests, production builds, recovery journeys, source-hashed evidence artefacts, and a repeatable latency benchmark.

The full-stack application also provides deployment and research infrastructure. Production configuration fails closed on weak secrets, origins, hosts, HTTPS, and rate limiting; health endpoints distinguish core readiness from optional model availability; Docker packaging, database persistence, backups, ownership enforcement, transient audio, and deterministic fallbacks are implemented rather than described only as future requirements. Separate consent, pseudonymous export, withdrawal, and dataset freezing remain available when the optional research workflow is used.

Finally, the evidence package and versioned configurations improve auditability. Model metrics, paired comparisons, policy fixtures, automated tests, performance summaries, smoke manifests, and protocol files identify which claim is supported by which artefact. Some private model-package hashes remain to be finalised, but the overall approach demonstrates how a production-oriented dissertation application can make its evidence boundaries inspectable.

6.5 Limitations of the Conclusions

6.5.1 Model and Data Limitations

The emotion-recognition conclusions derive from IEMOCAP, a corpus of ten actors completing acted or improvised English interactions. The dataset does not represent the full variation of spontaneous emotion, language, accent, age, disability, culture, device quality, or background noise that AffectLab may encounter. The four-class mapping simplifies expressive categories, and the exploratory mapping from fear to anxiety is especially weak. One primary training seed limits the assessment of optimisation variability.

Speaker-independent session folds reduce identity leakage but do not constitute external validation. Calibration can also deteriorate under domain shift even when it is strong on held-out sessions. The global packaging-chain summary adds a further limitation: its calibration fit and descriptive evaluation reuse all 5,531 utterance identities and labels in different fold roles, so it is not an independent held-out estimate. The final full-data packages likewise have no independent test set because all eligible benchmark examples are used for training after configuration selection. Their operational thresholds must therefore be treated as design parameters, not validated psychological boundaries.

6.5.2 Application and Measurement Limitations

The dialogue policies cover three bounded scenarios and cannot represent the ambiguity, power imbalance, culture, history, or stakes of all interpersonal situations. Feature extraction is transparent but lexically limited: paraphrases can be missed, indirect speech can be misread, and detected phrases do not establish communication competence. Generative responses remain dependent on an external provider when enabled, while template fallbacks trade linguistic variety for predictability.

The separately implemented cognitive-pattern, intent, possible-cause, readiness, and resistance fields are lexical support-routing heuristics, despite terminology that may imply psychological assessment. Their fixed rules have no construct- or clinical-validity evidence, and the present tests establish only selected input–output behaviour. The pilot did not isolate the feature or evaluate whether its labels and strategy choices were helpful, inaccurate, stigmatising, or harmful. The dissertation therefore makes no claim that these fields reveal cognition, identify a true cause or intention, quantify readiness or resistance, or select an optimal intervention.

Engineering evaluation used authored cases, deterministic services, synthetic accounts, and a small number of integrated journeys. The local latency benchmark used an in-process transport and memory repository. These establish behaviour and regression performance on the tested paths but not prolonged concurrent capacity, internet latency, deployed MongoDB latency, trained-inference or provider behaviour, adversarial traffic, diverse microphones, or every assistive technology. The present evidence therefore supports implementation completeness, not production readiness. Even a controlled deployment requires target-environment validation, independent security review, monitoring, incident response, secret management, and jurisdiction-specific governance.

6.5.3 Supporting User-Evidence Limitations

The supplementary pilot included 15 enrolled users, 12 of whom completed all three required tasks. Required tasks bypassed multimodal inference and affect-aware pacing, so the pilot did not evaluate whether the emotion-aware adaptation was acceptable, useful, accurate, intrusive, or beneficial. It also had no randomised control condition and used study-specific ordinal items. Incomplete tasks, missing ratings, demand characteristics, repetition, familiarity, and self-selection may all influence the summaries. The pilot therefore informs refinement of the baseline workflow but does not establish sustained skill acquisition, causal benefit, or transfer to real conversations. AffectLab is not a diagnostic, clinical, crisis, or therapeutic system.

6.6 Future Work

Future work should proceed in an order that strengthens the weakest evidence before increasing system autonomy. Table 6.1 summarises the proposed sequence.

Table 6.1: Prioritised future-work roadmap.

Priority	**Work package**	**Evidence produced**
Immediate	Staging performance, load, and resilience evaluation	Response-time distributions, throughput, saturation points, recovery time, and resource use with production persistence, models, providers, and expected concurrency.
Near term	External and stress-condition model evaluation	Cross-corpus classification, calibration under shift, device and noise robustness, and appropriately powered subgroup analyses.
Near term	Validate dialogue features and feedback with human annotation	Construct validity, inter-rater agreement, paraphrase coverage, error categories, and revised evidence rules.
Near term	Reframe or validate cognitive support-routing labels	Decision on removal or plain-language renaming, expert and user review, false-positive and false-negative cases, and comparison with a neutral routing baseline.
Medium term	Controlled comparison of baseline and affect-aware interaction	Supporting evidence about whether bounded adaptation changes pacing preference, task completion, perceived helpfulness, or intrusiveness.
Longer term	Longitudinal, multilingual, and participatory accessibility work	Retention, transfer, broader inclusion, privacy, and governance evidence.

6.6.1 Validate Deployment Performance and Reliability

The immediate next step is to repeat the local performance check in a staging environment that matches the intended deployment. This should use the production Nginx–FastAPI–MongoDB path, packaged text and audio models, configured external providers, realistic network conditions, and representative session histories.

Warm and cold starts, deterministic fallback turns, multimodal inference, transcription, replay, export, and branch creation should be measured separately so that a slow optional service is not hidden inside an aggregate average.

A concurrent-load tool should report throughput, median and tail latency, error rate, queue time, CPU, memory, and database utilisation as client count increases. Service-level objectives should be declared before the run, and tests should identify the saturation point rather than merely recording one successful request. Soak testing, forced provider timeouts, database restart, backup restoration, and process replacement should additionally measure recovery behaviour and verify that incomplete requests do not create duplicate turns or corrupt session versions.

Browser performance should be measured on the deployed build using Core Web Vitals and interaction traces on desktop and representative mobile hardware. These staging results would complement the current responsive-layout and accessibility suite and turn the existing deployment-oriented architecture into deployment-specific operational evidence.

6.6.2 Broaden Model Validation and Improve Uncertainty Handling

The model pipeline should be tested across corpora and, with explicit consent, on data closer to the intended use context. Cross-corpus evaluation would reveal how much of the IEMOCAP result depends on actors, recording conditions, and label conventions. Stress tests should vary microphones, compression, background noise, transcript errors, utterance length, and missing modalities. Multilingual work should begin with a new construct and annotation review rather than directly transferring the English label mapping.

Repeated training seeds and stronger baselines would better quantify optimisation uncertainty. Future comparisons could investigate longer or learned context selection, speaker-aware aggregation, modality-specific missingness training, and alternative late-fusion rules. These additions should retain paired evaluation and should not tune headline claims on the test partition.

Calibration requires continued attention after deployment. Reliability should be monitored by source, device condition, and data period where lawful and statistically justified. Selective prediction, abstention, or distribution-shift indicators may be more appropriate than forcing every input into an affect label. Thresholds should be revised only with held-out or prospectively collected evidence, and the application should preserve an accessible baseline regardless of model availability.

6.6.3 Evaluate and Refine the Adaptation Policy

The existing policy is intentionally small and deterministic. Future evaluation should ablate its inputs to determine whether confidence, modality agreement, and explicit preference each contribute useful information. Logs can identify how often actions are selected, declined, reversed, or followed by manual pacing changes. Qualitative interviews could reveal whether acknowledgements feel supportive, inaccurate, intrusive, or distracting.

Personalisation should remain transparent and opt-in. A user might prefer fewer affect acknowledgements, text-only operation, slower pacing, or different uncer-

tainty wording. Such preferences can be stored as explicit controls rather than inferred as stable psychological traits. Any learned policy should first be evaluated offline against safety and autonomy constraints and compared with the deterministic baseline. Reinforcement learning based on engagement alone would be inappropriate because longer interaction is not equivalent to better rehearsal.

6.6.4 Validate Feedback and Communication Measures

The deterministic feature rules should be compared with independent human annotations of the same turns. This would quantify precision, recall, agreement, synonym coverage, and systematic error for features such as a clear request, boundary maintenance, non-blaming language, and a specific need. The analysis should include indirect phrasing, negation, reported speech, culturally varied wording, and speech-to-text errors.

The cognitive support-routing component should be treated as a separate decision point rather than silently included in that validation. One option is removal or replacement of psychological terms with directly observable descriptions such as "asks for steps", "contains a causal connector", or "pushes back on a suggestion". If the named constructs are retained, the work would require operational definitions, appropriate domain expertise and ethics review, independently annotated and representative cases, participant correction mechanisms, error and subgroup analysis, and a comparison showing that construct-labelled routing adds benefit over a neutral intent menu or clarification question. Clinical wording or use should remain out of scope unless supported by an appropriately governed clinical study.

Semantic or learned detectors could improve paraphrase coverage, but they should return evidence spans and uncertainty rather than opaque trait scores. Feedback usefulness should also be evaluated separately from detector accuracy: a technically correct observation may still be poorly timed or unhelpful. Co-design with communication educators and target users could refine wording while preserving the distinction between observable behaviour and personal judgement.

6.6.5 Study Transfer, Accessibility, and Long-Term Use

Later studies should examine whether users return voluntarily, whether repeated practice remains useful, and whether any measured change persists. Transfer to real conversations would require independent outcomes, such as user-reported application of a planned request or externally rated role-plays, not merely improvement on the same scenario. Such studies should avoid clinical claims unless they adopt an appropriate clinical design and oversight.

Accessibility evaluation should involve users of keyboards, screen readers, speech input, captions, reduced-motion settings, and different cognitive or communication needs. The current interface and automated checks provide a foundation but cannot replace participatory accessibility testing. Broader language and cultural adaptation should include local review of scenario norms, refusal styles, power relationships, consent materials, and safety resources.

6.6.6 Strengthen Deployment, Privacy, and Governance

Before any controlled deployment, the system should complete staging load tests, an independent security review, production backup-and-recovery exercises, dependency and model monitoring, operational alerting, and documented incident response. Validation should include the intended MongoDB service, trained inference, configured providers, real network transit, expected concurrency, and adversarial traffic. Provider changes should be treated as versioned operational factors because they can alter transcription, generation, latency, or failure modes. Model and policy artefacts should be signed or hashed, and the remaining private-package provenance gaps should be closed.

Data minimisation should continue to guide development. Audio should remain transient unless a separately justified protocol requires retention, and derived research records should avoid reconstructable conversation content. Retention, deletion, export access, and withdrawal should be tested periodically in the deployed environment. Governance should also define who may change scenarios, thresholds, prompts, safety resources, or analysis code and how those changes are reviewed and recorded.

6.7 Final Conclusion

This dissertation shows that multimodal affect recognition can be connected to conversational role-play without making the classifier the authority over the interaction. Causal dialogue context improved textual emotion classification on the chosen benchmark, and audio supplied complementary information that improved fused classification. The accompanying calibration results demonstrated that stronger classification does not automatically produce trustworthy confidence. This finding shaped the application design: uncertainty is retained, adaptation is bounded, users can decline it, and missing inference preserves the core workflow.

AffectLab's broader contribution is therefore practical, architectural, and methodological as much as predictive. Deterministic scenario progression, evidence-grounded feedback, constrained generation, safety precedence, provenance, and complementary verification layers make the system's behaviour more inspectable. The completed engineering evidence establishes that this design works end to end across its tested policies, interfaces, persistence, accessibility, recovery, and local-response paths. It is a helpful communication-practice application, not merely a model demonstration.

The completed system is implementation-complete and suitable for controlled deployment after environment-specific validation. Its present evidence covers production builds and configuration checks, health and recovery paths, security and ownership controls, deterministic fallbacks, broad automated coverage, real-database smoke journeys, desktop/mobile accessibility checks, and local responsiveness for the offline deterministic core. It does not cover live trained inference or providers, real network transit, deployed MongoDB latency, representative concurrency, adversarial testing, or independent security review, and private model-file provenance remains incomplete. Those conditions must be validated in the target environment before a production-readiness claim is made. Likewise, benchmark performance does not establish emotional truth, and the small uncontrolled pilot does

not establish participant benefit. Within those boundaries, the dissertation delivers a thoroughly developer-tested and reproducibly evaluated application for emotion-aware communication rehearsal while preserving uncertainty and user agency.

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Appendix A

Generative AI Tools Usage

Generative AI tools were used as supplementary resources during the development and documentation of this thesis. Their use was limited to supporting literature exploration, improving the clarity of academic writing, assisting with LaTeX formatting, and providing guidance during technical problem-solving. All generated suggestions were reviewed before being incorporated into the thesis.

An overview of the generative AI tools used is presented in Table A.1.

AI Tool	Use Case	Used in	Remarks
ChatGPT 5.6 Sol / 6 Astra	Literature exploration and identification of relevant scientific publications	Chapter 2	Suggested references were independently verified
	Improvement of academic phrasing, grammar, and text clarity	Chapters 1–6	Suggestions were reviewed and adapted
	Assistance with LaTeX syntax, formatting, and document structure	Entire thesis	Formatting suggestions were manually checked
	Technical guidance and debugging suggestions	Implementation chapters	Suggestions were evaluated and tested before use

Table A.1: Overview of generative AI usage

A.1 ChatGPT Details

ChatGPT (versions 5.6 Sol and 6 Astra, developed by OpenAI) was used as a supplementary assistant throughout selected stages of the thesis. Its contributions are described below.

A.1.1 Literature Exploration

Purpose: To support the identification of relevant scientific publications and research directions related to the topics investigated in this thesis.

Method: ChatGPT was prompted to suggest academic publications, relevant research topics, and potential directions for organizing the state-of-the-art review in Chapter 2.

Verification: Suggested publications were checked against academic sources to verify their existence, relevance, and bibliographic information. The original publications were consulted before being referenced.

Implementation: The verified references and research directions were used as supporting resources when developing the literature review.

A.1.2 Academic Writing Assistance

Purpose: To improve the readability, grammatical correctness, and academic style of selected passages.

Method: Selected paragraphs were provided to ChatGPT with requests to improve sentence structure, clarity, or academic phrasing while preserving their original meaning.

Verification: The suggested revisions were reviewed to ensure that the technical meaning and intended interpretation remained unchanged.

Implementation: Selected linguistic improvements were incorporated into the thesis after manual revision.

A.1.3 LaTeX Formatting Assistance

Purpose: To assist with document formatting and the resolution of LaTeX-related issues.

Method: ChatGPT was consulted for guidance on LaTeX commands, table formatting, document structure, and compilation errors.

Verification: Suggested solutions were checked by compiling the document and inspecting the resulting output.

Implementation: Validated formatting solutions were applied where appropriate.

A.1.4 Technical Guidance and Debugging

Purpose: To obtain supplementary explanations and troubleshooting suggestions during the implementation process.

Method: ChatGPT was occasionally consulted to clarify technical concepts, explain error messages, and suggest possible approaches to resolving implementation difficulties.

Verification: Suggestions were assessed against the project requirements and, where applicable, tested in the development environment.

Implementation: Only solutions considered appropriate and successfully validated were incorporated into the implementation.

Generative AI was used exclusively as a supporting tool. The selection of research methods, system design decisions, implementation validation, interpretation

of experimental results, and final conclusions remained the responsibility of the author.